

Supporting Information

Pore-Space-Partition-Oriented Sandwich Platinum Array Confined in a Metal-Organic Framework for Boosting Overall Water Splitting

Jia-Min Huo,^{†,§} Ying Wang,^{*,†} Ze-Lin Ma,[†] Jie Meng,[†] Tian-Shu He,[†] Bo-Lin Sun,[†] Shu-Ni Li,^{*,†}
and Quan-Guo Zhai^{*,†}

[†]Key Laboratory of Applied Surface and Colloid Chemistry, Ministry of Education, Key Laboratory of Macromolecular Science of Shaanxi Province, School of Chemistry & Chemical Engineering, Shaanxi Normal University, Xi'an, Shaanxi 710062, China

[‡]State Key Laboratory of Solidification Processing, Center for Nano Energy Materials, School of Materials Science and Engineering, Northwestern Polytechnical University and Shaanxi Joint Laboratory of Graphene (NPU), Xi'an, Shaanxi 710072, China

[§]School of Chemistry and Chemical Engineering, Yulin University, Yulin, Shaanxi 719000, China

S1. Experimental Details

S1.1. Materials and reagents

Chemical reagents and materials, including $\text{Fe}(\text{NO}_3)_3$, $\text{Ni}(\text{NO}_3)_2$, bicyclo[1.1.1]pentane (BCP), 1,4-terephthalic acid (H_2BDC), naphthalene-2,6-dicarboxylic acid (2,6-NDC), K_2PtCl_4 , N,N-dimethylacetamide (DMA), 1,3-dimethyl-3,4,5,6-tetrahydro-2(1H)-pyrimidinone (DMPU), 1,1,1,5,5,5-hexafluoro-2,4-pentanedione (HFP), CH_3OH and HCOOK were purchased from Aladdin and China National Pharmaceutical Group Corporation (Shanghai, China). Deionized water was prepared in the lab.

S1.2. Characterizations

Powder X-ray diffractions (PXRD) were carried out on a MiniFlex600 X-ray diffractometer (Rigaku, Japan) using a Cu-K α source equipped within the range of $2\theta = 5\text{--}50^\circ$. The morphological and element mapping of as-prepared samples were measured on scanning electron microscope (SEM, SU8220, HITACHI, Japan) and transmission electron microscope (TEM, JEM-2800F, JEOL, Japan). Energy-dispersive X-ray analyses (EDX) were tested on by using a Quanta 200 spectrometer (FEI Co. Ltd, America). Chemical composition and state of samples were investigated using the X-ray photoelectron spectroscopy (XPS, AXIS ULTRA, Japan). All of the binding energy was corrected using the C 1s peak as an internal standard at 284.6 eV except for Pt. Inductively coupled plasma optical emission spectroscopy (ICP-OES) spectrometry of samples was tested on AGILENT ICP 730 (Agilent Co. Ltd, America). N_2 adsorption isotherms were carried out on a Micromeritics 3-Flex surface-area and pore-size analyzer at standard atmospheric pressure by the static volumetric method. Acetonitrile-exchanged compounds were loaded into the sample tube and dried at 80 °C to remove the solvent. The X-ray absorption spectra of Fe, Ni and Pt including X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) of the samples were conducted in fluorescence mode at the BL14W1 beamline of the Shanghai Synchrotron Radiation Facility, China, operated at 3.5 GeV with maximum injection currents of 230 mA. The synchrotron beam was monochromatized using a double-crystal monochromator equipped with a Si (111) crystal to reduce the harmonic component of the monochrome beam.

Electrochemical measurements

All electrochemical measurements were carried out in 0.5 M H₂SO₄ and 1 M KOH (pH = 13.7) solution on an electrochemical workstation (CHI 760E) with a three-electrode system. The prepared samples were used as the working electrodes and a graphite rod and Hg₂Cl₂/Hg (HgO/Hg) electrode in 0.5 M H₂SO₄ (1 M KOH) were used as the counter and reference electrode, respectively. The as-prepared catalyst (10 mg) and acetylene black (10 mg) was added into 1 mL of mixture solution of water (640 μ L), ethanol (310 μ L) and Nafion (50 μ L) and sonicated for at least 1 h forming ink to prepare electrocatalyst. The ink (150 μ L) was dropped on the cleaned carbon cloth for HER, OER and water splitting. The working area of the carbon cloth electrode was 1 cm² and the catalyst loading amount was 1.5 mg·cm⁻². Before LSV measurements, cyclic voltammetry curves (CV) were used to activate the working electrode. Linear sweep voltammetry (LSV) measurements were tested at a scan rate of 5 mV·s⁻¹ with 95 % *iR*-compensation. Electrochemical impedance spectroscopy (EIS) was carried out by applying with 5 mV amplitude in a frequency range from 0.1 Hz to 100 kHz. Electrochemically active surface areas (ECSA) were evaluated by measuring the double-layer capacitance (*C*_{dl}) via cyclic voltammograms (CV) in a potential range of 1.1 to 1.2 V vs. RHE for OER in 1 M KOH, 0.3 to 0.4 V vs. RHE for HER in 0.5 M H₂SO₄ and 0.4 to 0.5 V vs. RHE for HER in 1 M KOH, in which without redox process by $C_{dl} = I/\nu$, where *I* was the charging current (mA·cm⁻²), and ν was the scan rate (mV·s⁻¹). The *C*_{dl} values were calculated by using the CV curves from 5 to 100 mV·s⁻¹ scan rates. The long-term durabilities of these samples were assessed by the *i-t* testing at the potential of $\eta_{100 \text{ mA}\cdot\text{cm}^{-2}}$ for 12 h in 0.5 M H₂SO₄ and 1 M KOH.

Faradaic efficiency measurements: The closed electrolyser was purged with N₂ for several time to completely remove the air in the system. In the N₂-saturated system, the generation H₂ over the Pt@MOF-BCP catalyst at 100 mA·cm⁻² was based on gas chromatography measurement (GC9790 II, Fuli, Zhejiang, China) every 20 min. Faradaic efficiency of Pt@MOF-BCP was calculated by the ratio of amount of H₂ experimentally generated to the theoretically calculated H₂.

Turnover frequencies (TOF) for HER and OER at different potentials were calculated as follow equation:

$$\text{TOF} = jS/(zFn)$$

where *j* (A·cm⁻²) is the current density, *S* is the surface area (1 cm²), *z* the number of electrons

transfer, F is the Faraday constant of $96485 \text{ C}\cdot\text{mol}^{-1}$, n is the total metal ions molar number calculated from the EDX results of Pt@MOFs. Assuming that the Ni, Fe and Pt sites were all catalytically active, the TOF values of Pt@MOFs and commercial Pt-based catalysts were calculated.

Metals mass activity was calculated as follow:

$$\text{mass activity} = j/m_{\text{metal}}$$

where $j \text{ (A}\cdot\text{cm}^{-2}\text{)}$ is the current density, m_{metal} is metal loading mass of catalyst from EDX. In order to compare the metals activities, their activity values have been normalized to all metals.

BET-normalized LSV curves were calculated as follow:

$$j_{\text{surface areas}} = j/\text{BET}$$

In ICP-OES experiments, Pt@MOFs powder (8.8 mg) was deconstructed by 1 mL nitric acid and 1 mL perchloric acid under 200°C with 1 h. Then the deconstructed-solution and electrolyte after the reaction were transferred to a 10 mL volumetric flask and filled to constant volume with H_2O and diluted the obtained solution tenfold. The diluted solution was test on ICP-OES equipment. The actual concentration of metal elements was calculated as:

$$C(\text{mg/kg}) = \frac{C_x(\text{mg/L}) * f * V(\text{mL}) * 0.001}{m(\text{g}) * 0.001}$$

where C_x is the concentration obtained from ICP-OES equipment, f is the dilution ratio, V is the constant volume, m is the sample mass.

S1.3. Synthesis

Preparation of MOF precursors. To prepare MOF-BCP, a mixture solution of $\text{Fe}(\text{NO}_3)_3$ (20 mg), $\text{Ni}(\text{NO}_3)_2$ (20 mg), BCP (31 mg), TPP (21 mg) and 3 g DMA/ H_2O ($v/v = 5/1$) was sealed in a 20 mL vial and heated at 130°C for 1 day. The precipitate was filtrated and washed with EtOH. The precipitate was placed in a vacuum oven at 60°C to remove the residual EtOH.

To prepare MOF-BDC, a mixture solution of $\text{Fe}(\text{NO}_3)_3$ (30 mg), $\text{Ni}(\text{NO}_3)_2$ (30 mg), BDC (33 mg), TPP (30 mg) and 3 g DMA/ H_2O ($v/v = 5/1$) was sealed in a 20 mL vial and heated at 130°C for 12 h. After cooling to room-temperature, the precipitate was centrifuged and washed with EtOH, respectively. The precipitate was placed in a vacuum oven at 60°C to remove the residual EtOH.

To prepare MOF-NDC, a solution mixture of $\text{Fe}(\text{NO}_3)_3$ (30 mg), $\text{Ni}(\text{NO}_3)_2$ (30 mg), NDC (43 mg), TPP (62 mg), 6 g DMA/DMPU ($v/v = 2/1$), and 28 μL HFP was sealed in a 20 mL vial and heated

at 130 °C for 1 day. After cooling to room-temperature, the precipitate was centrifuged and washed with EtOH, respectively. The precipitate was placed in a vacuum oven at 60 °C to remove the residual EtOH.

Preparation of MIL-88. FeCl₃ (0.13 g), Ni(NO₃)₂ (0.13 g), BDC (0.1661 g) and DMF (10 mL) were sealed in a 20 mL vial and heated at 100 °C for 12 h. After cooling to room-temperature, the precipitate was centrifuged and washed with EtOH, respectively. The precipitate was placed in a vacuum oven at 60 °C to remove the residual EtOH.

Preparation of Pt@MOFs. MOF precursors (20 mg) and K₂PtCl₄ (10 mg) were added into 10 ml H₂O. The mixed solution was ultrasonically dispersed for 10 seconds and heated at 90 °C (for 8 h to MOF-BCP precursor, 12 h for MOF-BDC precursor and MOF-NDC precursor). These products were centrifuged and washed with EtOH and then were placed in a vacuum oven at 60 °C to remove the residual EtOH. Then 20 mg precipitates and HCOOK (5 mg) were added into 5 mL CH₃OH and 100 μ L H₂O. The mixed solutions were heated at 110 °C for 6 h (90 °C for MOF-BCP precursor). These products were second centrifuged and washed with EtOH and then were placed in a vacuum oven at 60 °C to remove the residual EtOH. Then the final products were obtained, and named as Pt@MOFs.

S1.4 Density Functional Theory (DFT) calculations

Density functional theory (DFT) calculations were performed using the DMol3 module within the Materials Studio software suite. The exchange-correlation interactions were described using the Perdew-Burke-Ernzerhof (PBE) functional, a widely adopted generalized gradient approximation (GGA) approach. Spin-polarized calculations were implemented with a Quintet numerical polarization basis set. The self-consistent field (SCF) procedure employed a convergence criterion of 1.0×10^{-6} Hartree (Ha) for electronic energy minimization. For geometry optimization, the convergence thresholds were set at 1.0×10^{-5} Ha for energy, 0.002 Ha/Å for maximum force, and 0.005 Å for maximum atomic displacement. To explicitly elucidate the spatial distribution and electronic interactions between Pt clusters and the MOF, Pt atoms were selected as a DFT-optimized model to approximate the electrochemical behavior of the Pt.

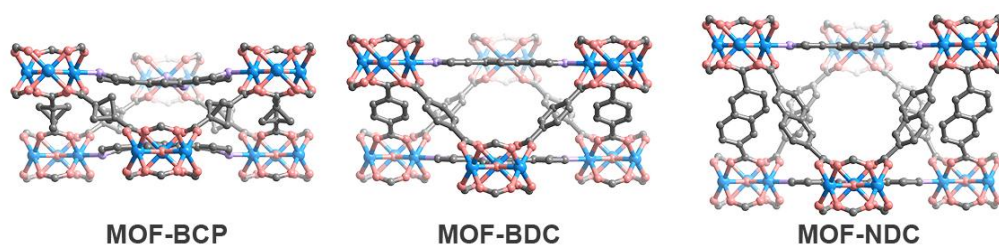


Figure S1. The simplified model of MOF precursors.

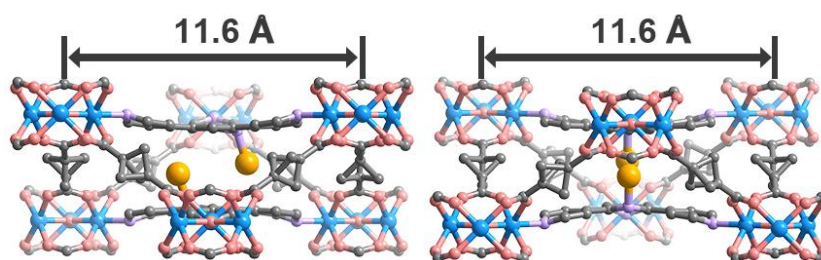


Figure S2. Computed position of Pt atoms in Pt@MOF-BCP.

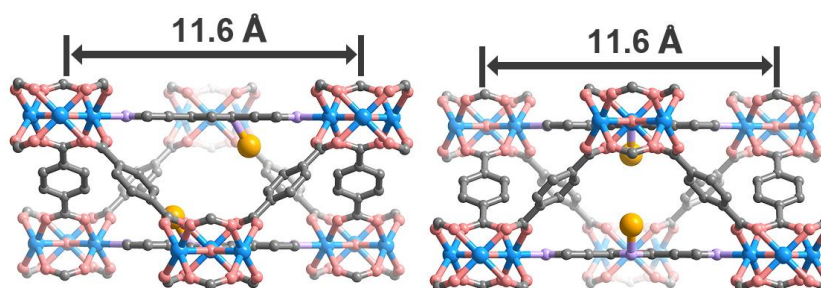


Figure S3. Computed position of Pt atoms in Pt@MOF-BDC.

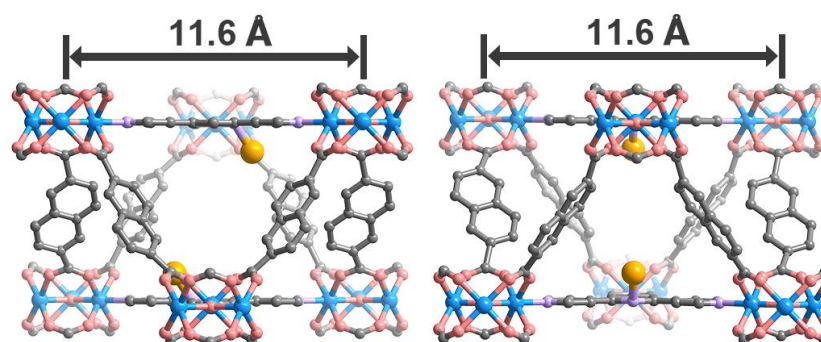


Figure S4. Computed position of Pt atoms in Pt@MOF-NDC.

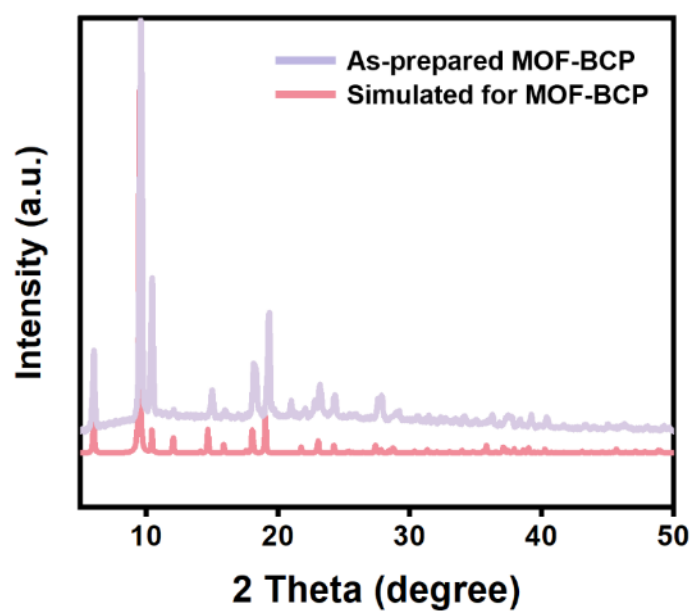


Figure S5. PXRD patterns of MOF-BCP precursor.

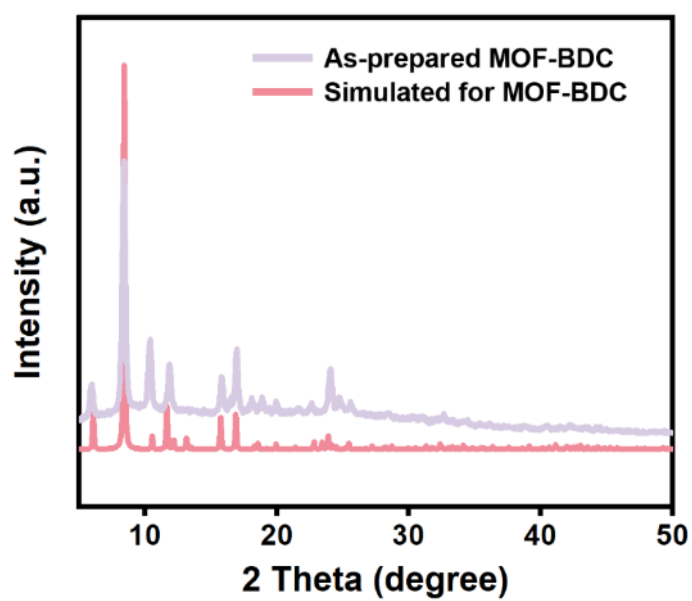


Figure S6. PXRD patterns of MOF-BDC precursor.

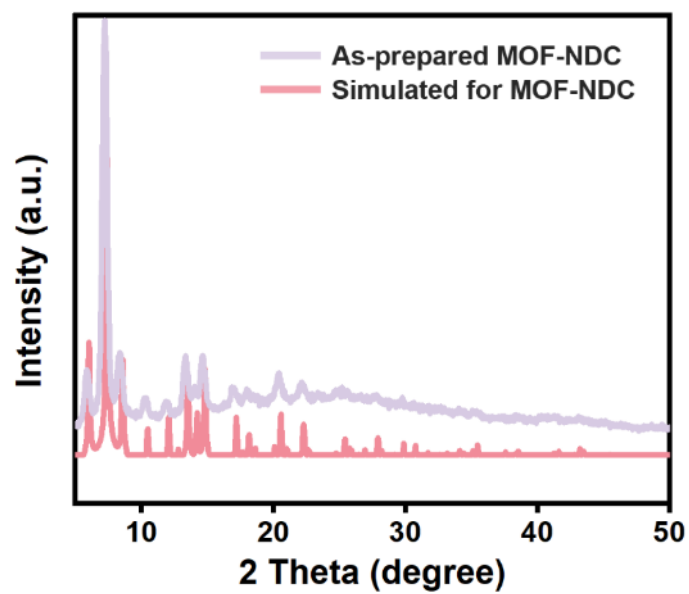


Figure S7. PXRD patterns of MOF-NDC precursor.

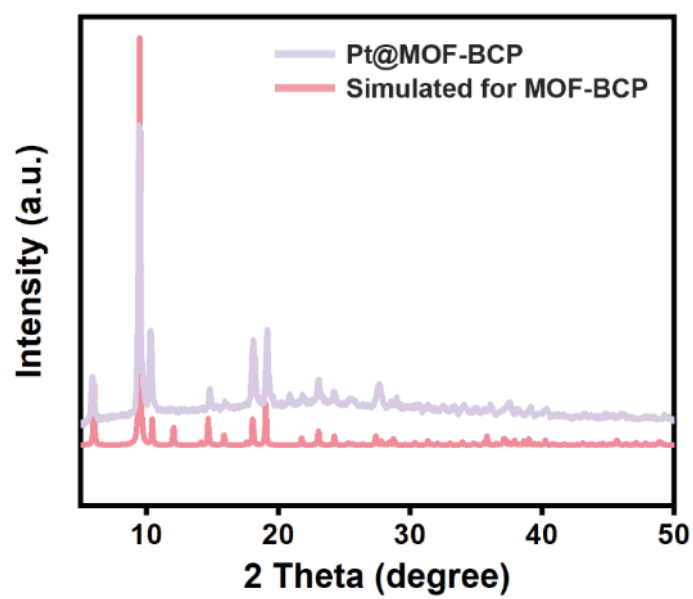


Figure S8. PXRD patterns of Pt@MOF-BCP.

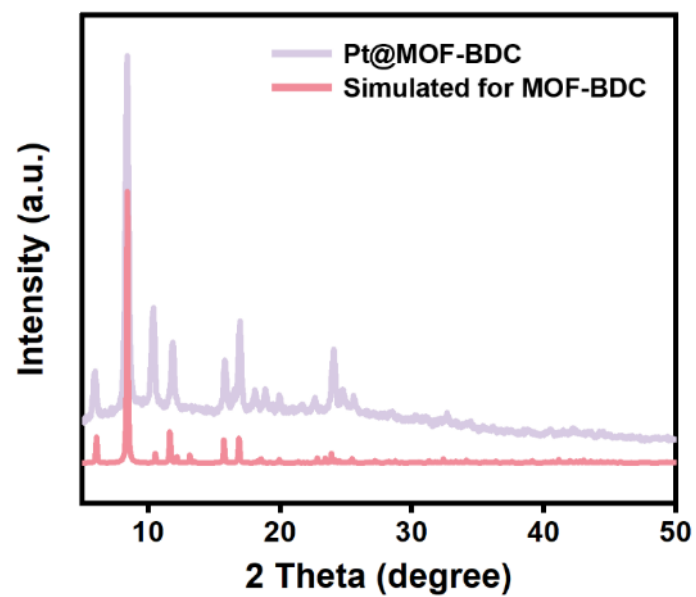


Figure S9. PXRD patterns of Pt@MOF-BDC.

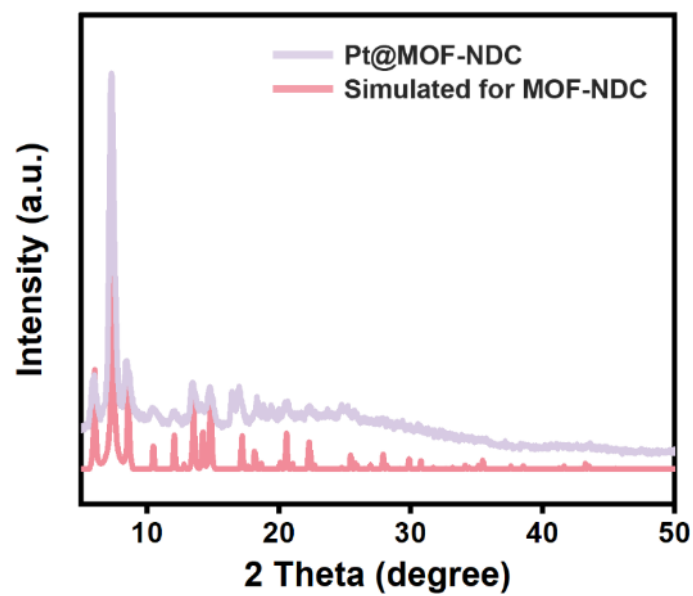


Figure S10. PXRD patterns of Pt@MOF-NDC.

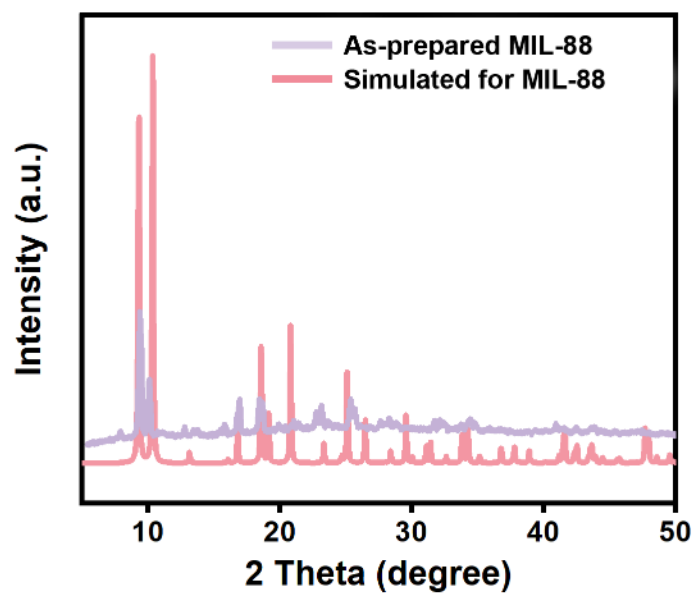


Figure S11. PXRD patterns of MIL-88 precursor.

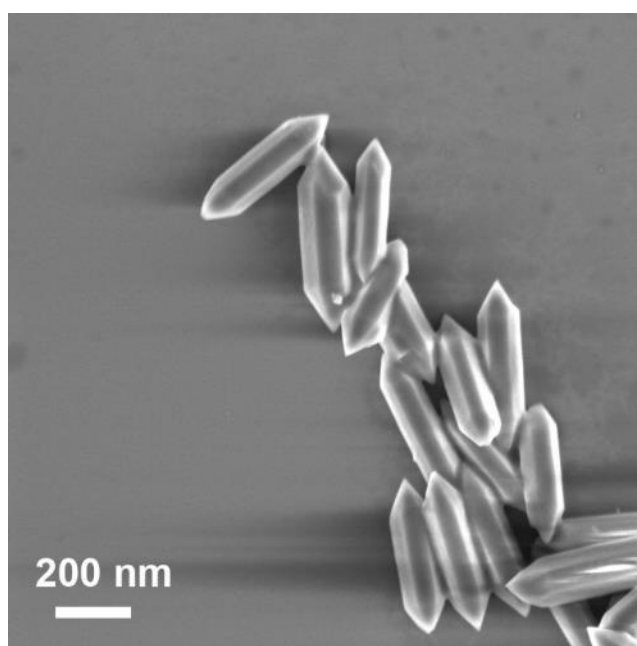


Figure S12. SEM image of MIL-88 precursor.

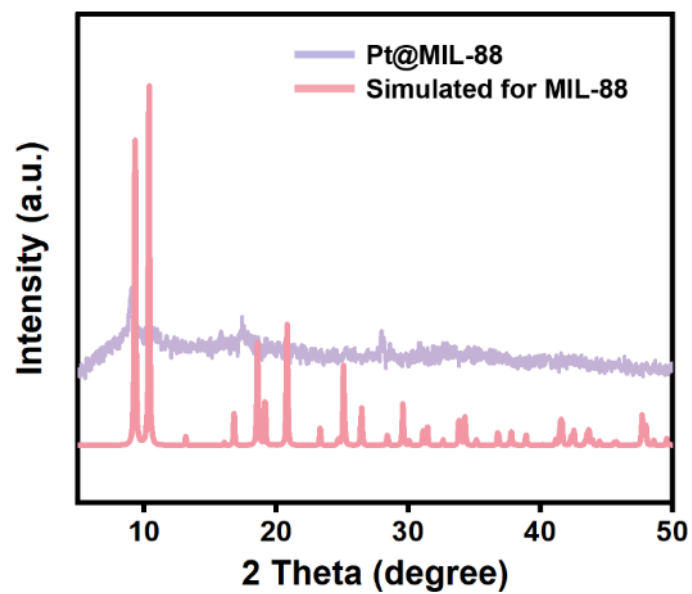


Figure S13. PXRD patterns of Pt@MIL-88.

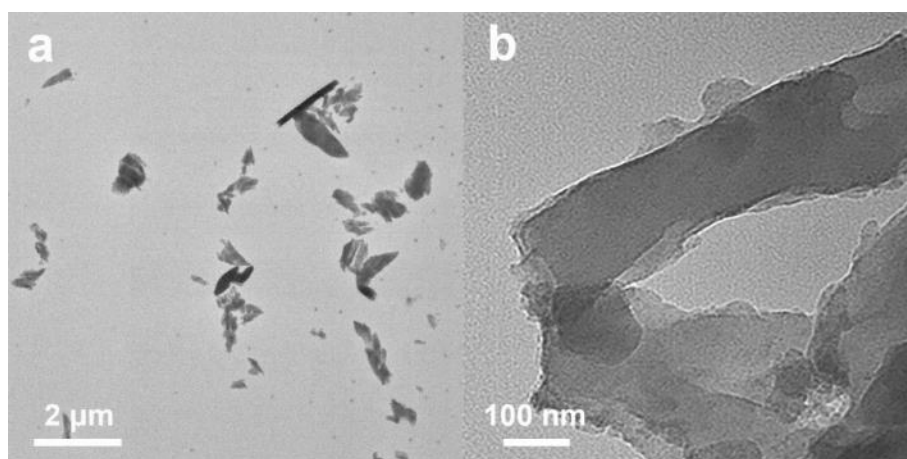


Figure S14. TEM images of Pt@MIL-88.

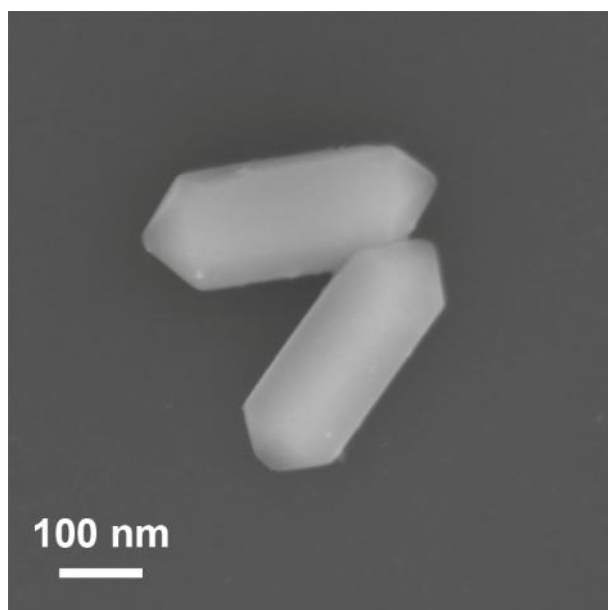


Figure S15. SEM image of MOF-BCP precursor.

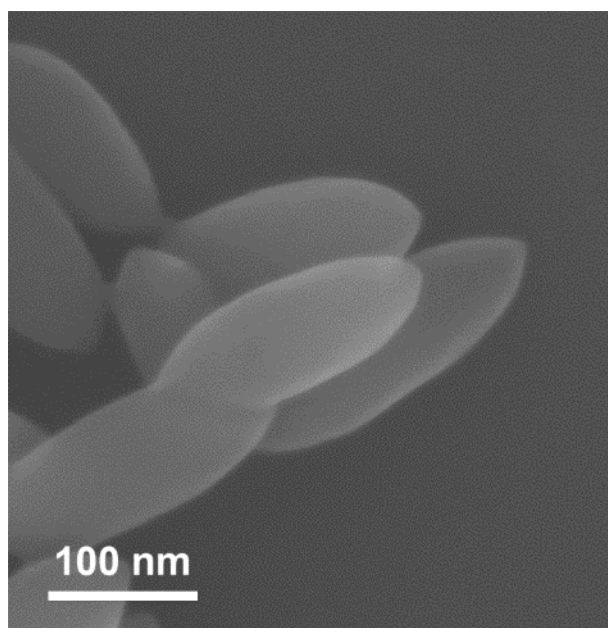


Figure S16. SEM image of MOF-BDC precursor.

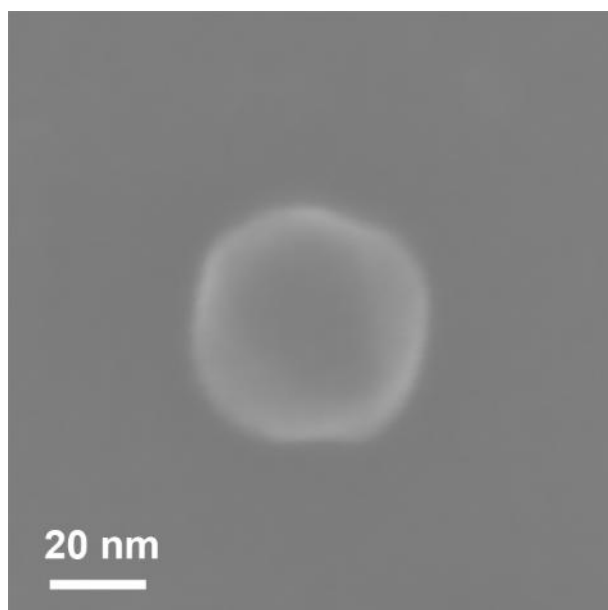


Figure S17. SEM image of MOF-NDC precursor.

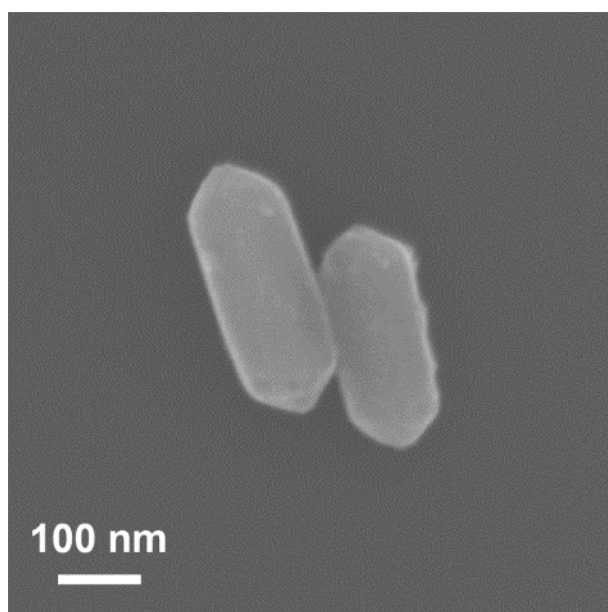


Figure S18. SEM image of Pt@MOF-BCP.

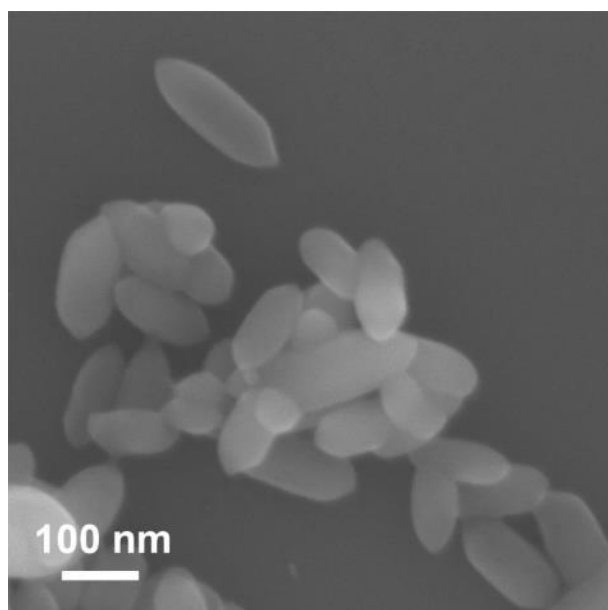


Figure S19. SEM image of Pt@MOF-BDC.

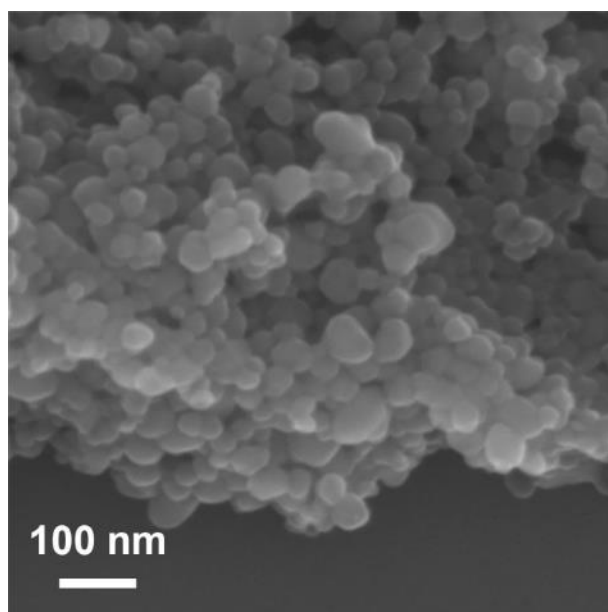


Figure S20. SEM image of Pt@MOF-NDC.

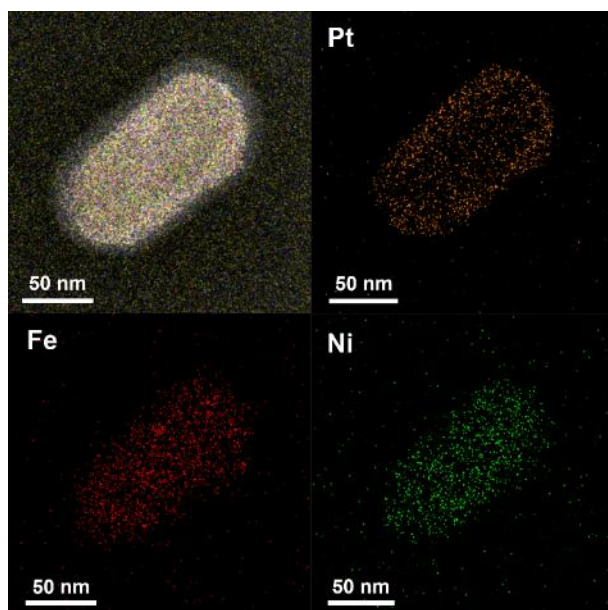


Figure S21. Mapping images of Pt@MOF-BCP.

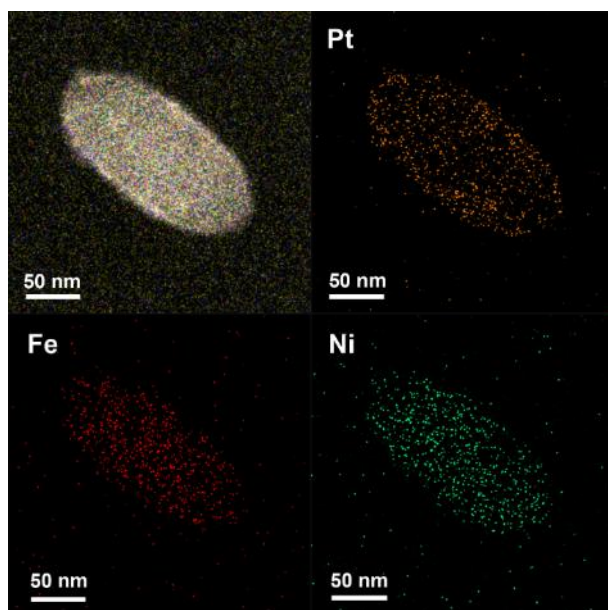


Figure S22. Mapping images of Pt@MOF-BDC.

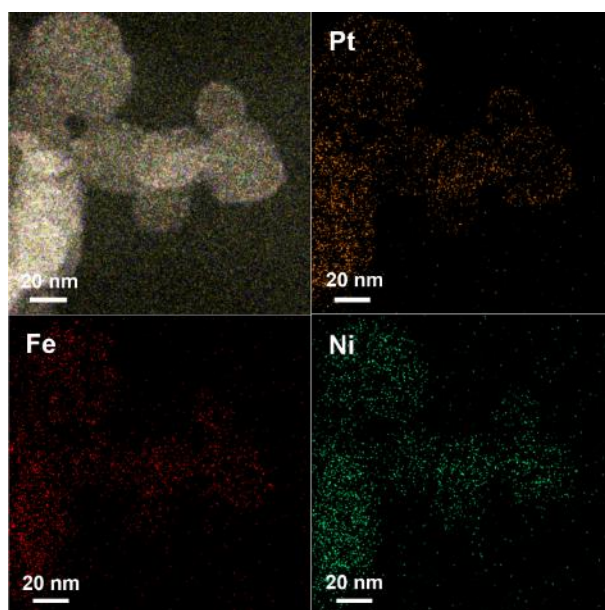


Figure S23. Mapping images of Pt@MOF-NDC.

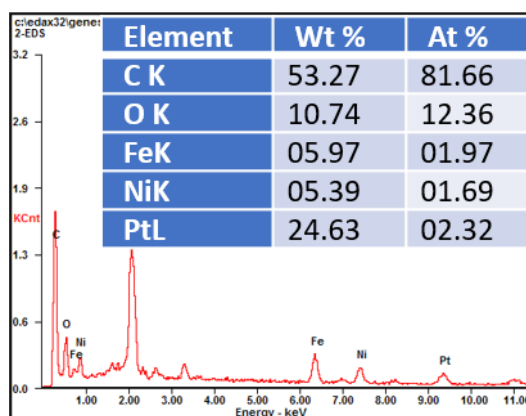


Figure S24. The EDX of Pt@MOF-BCP.

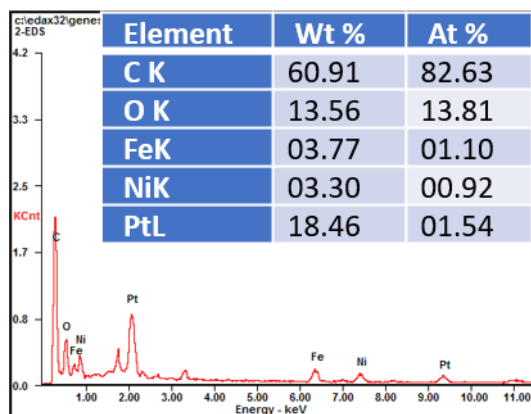


Figure S25. The EDX of Pt@MOF-BDC.

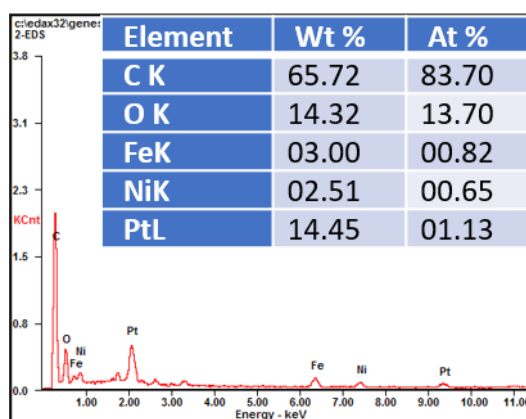


Figure S26. The EDX of Pt@MOF-NDC.

Table S1. The content of Pt, Fe and Ni of Pt@MOFs determined by ICP-OES.

Sample	Fe (wt %)	Ni (wt %)	Pt (wt %)
Pt@MOF-NDC	5.1	5.9	8.2
Pt@MOF-BDC	4.5	4.8	16.0
Pt@MOF-BCP	4.2	4.0	22.1

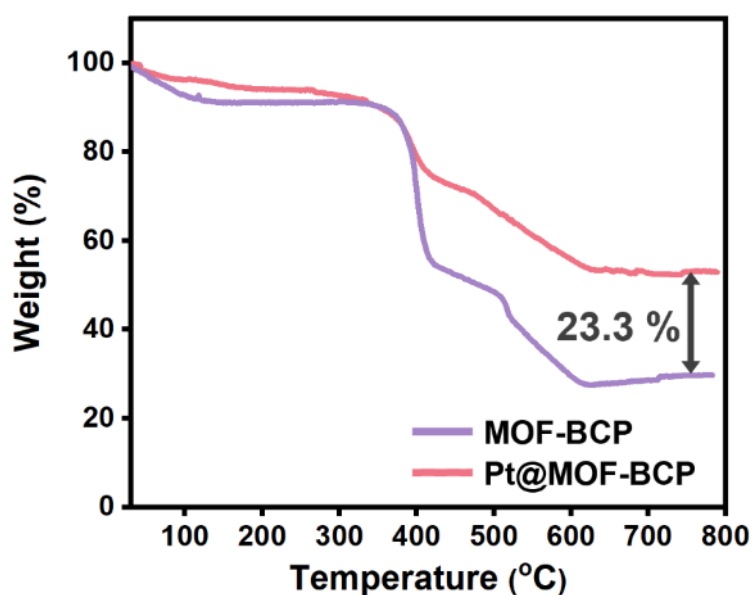


Figure S27. Thermogravimetric analyses (TGA) curves for the MOF-BCP and Pt@MOF-BCP.

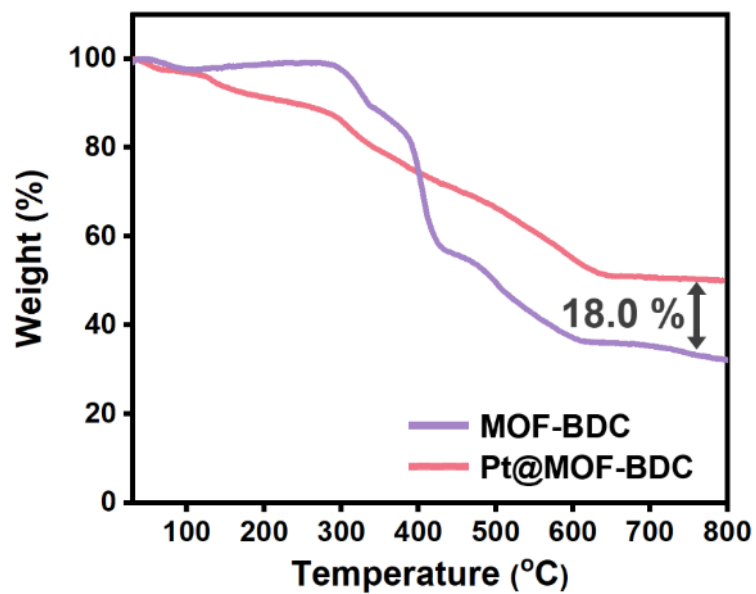


Figure S28. Thermogravimetric analyses (TGA) curves for the MOF-BDC and Pt@MOF-BDC.

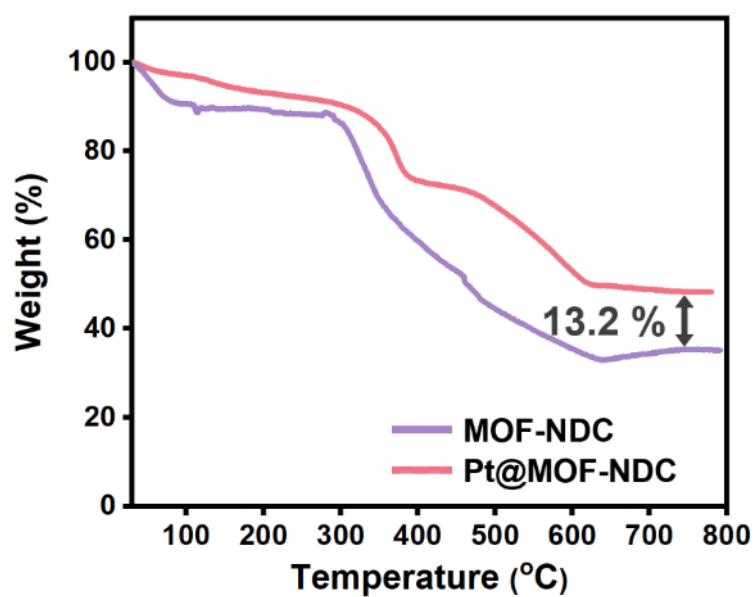


Figure S29. Thermogravimetric analyses (TGA) curves for the MOF-NDC and Pt@MOF-NDC.

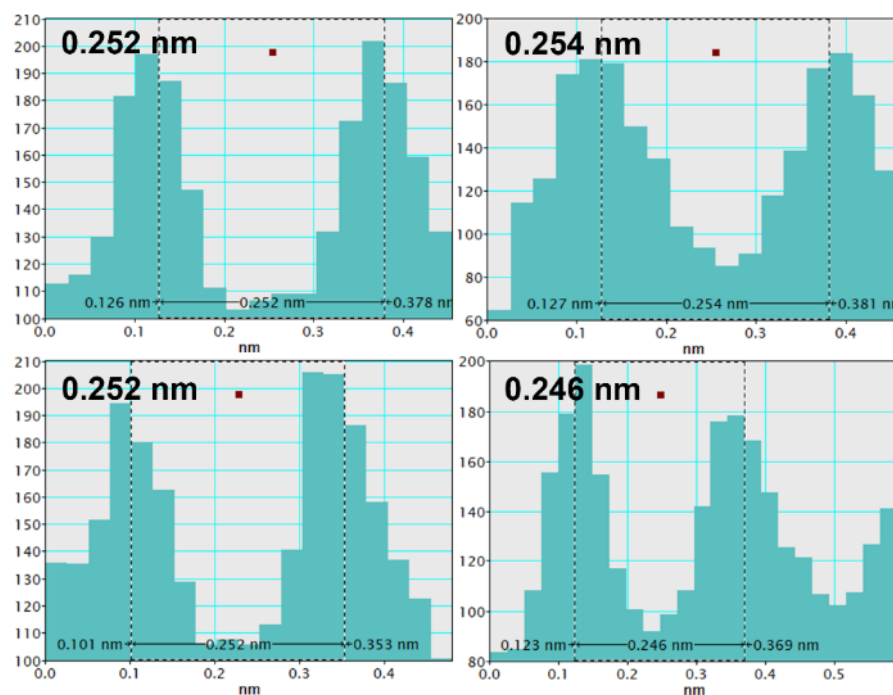


Figure S30. The lattice spacings of Pt@MOF-BCP.

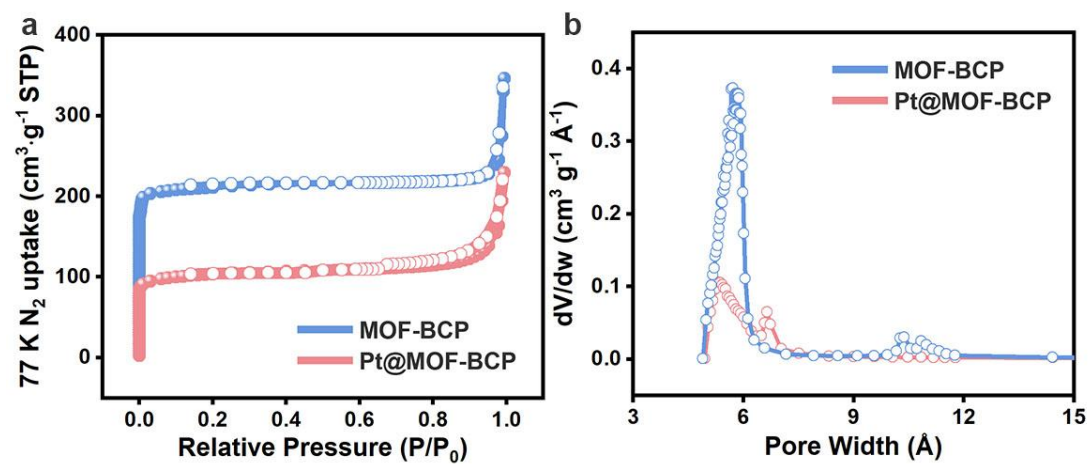


Figure S31. N_2 isotherms at 77 K and pore size distribution of MOF-BCP and Pt@MOF-BCP.

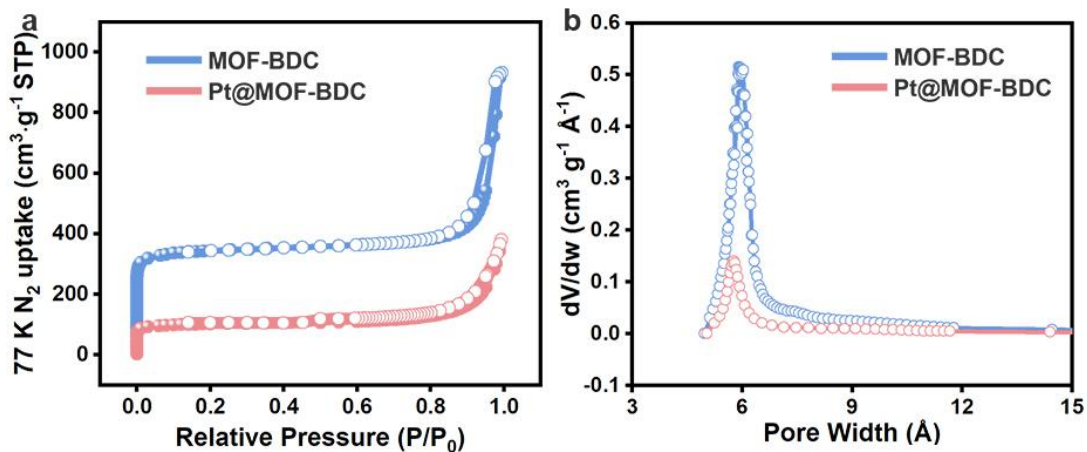


Figure S32. N_2 isotherms at 77 K and pore size distribution of MOF-BDC and Pt@MOF-BDC.

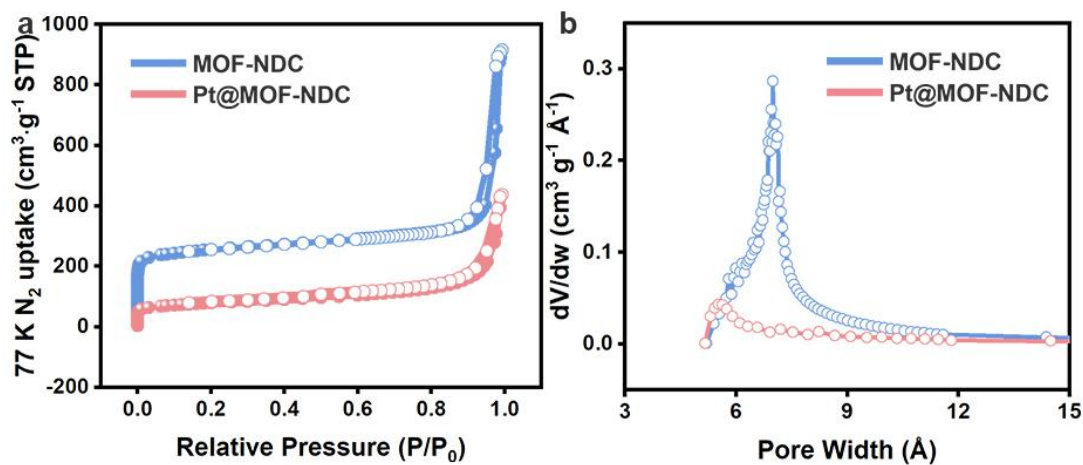


Figure S33. N_2 isotherms at 77 K and pore size distribution of MOF-NDC and Pt@MOF-NDC.

Table S2. The comparison of pore structural parameters of Pt@MOFs and corresponding MOF precursors.

Sample	S_{BET} ($\text{m}^2 \cdot \text{g}^{-1}$)	Pore size ($\text{\AA}$)	Pore Volume ($\text{cm}^3 \cdot \text{g}^{-1}$)
Pt@MOF-BCP	393.4	5.2	0.34
MOF-BCP	820.4	5.4	0.52
Pt@MOF-BDC	390.0	5.6	0.57
MOF-BDC	1325.6	5.7	1.44
Pt@MOF-NDC	281.2	5.6	0.66
MOF-NDC	960.2	7.0	1.40

Note: Detailed N_2 adsorption-desorption isotherms confirm that MOF-BDC exhibits the largest specific surface area ($1325.6 \text{ m}^2 \cdot \text{g}^{-1}$) compared with MOF-BCP and MOF-NDC. Reasonably, MOF-BCP, constructed with the shortest ditopic linker displays the smallest surface area and pore volume. MOF-NDC, which coordinated with the longest length ditopic linker ($9.1 \text{ \AA}$), exhibits the intermediate-values of surface area and pore volume unexpectedly. The excessive ditopic linker of NDC leads to suboptimal pore architecture and partial incomplete coordination, diminishing the porosity advantageous for N_2 adsorption.

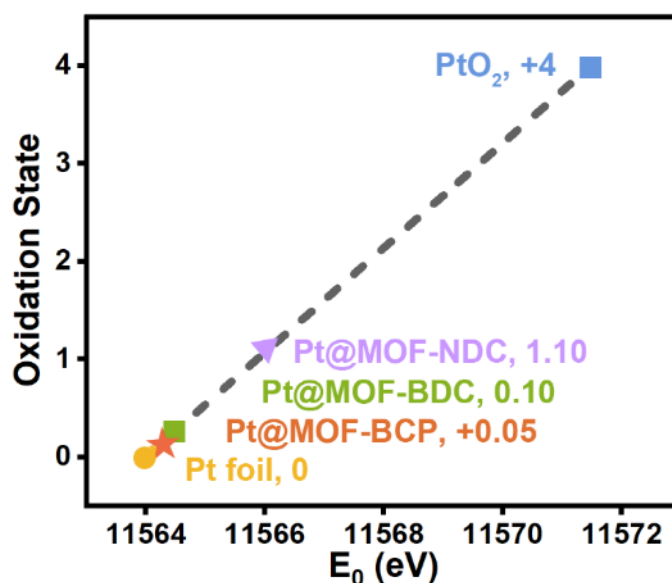


Figure S34. The calculated Pt oxidation state derived from XANES spectrum.

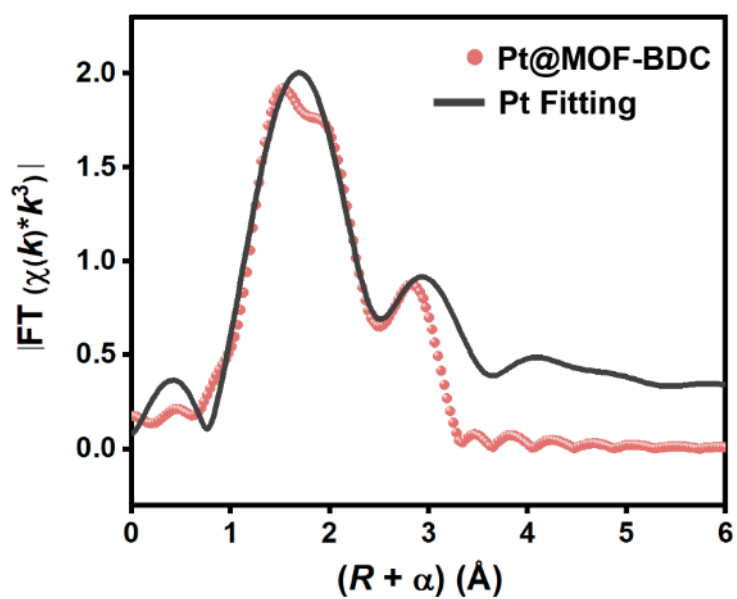


Figure S35. EXAFS fitting results of Pt for Pt@MOF-BDC.

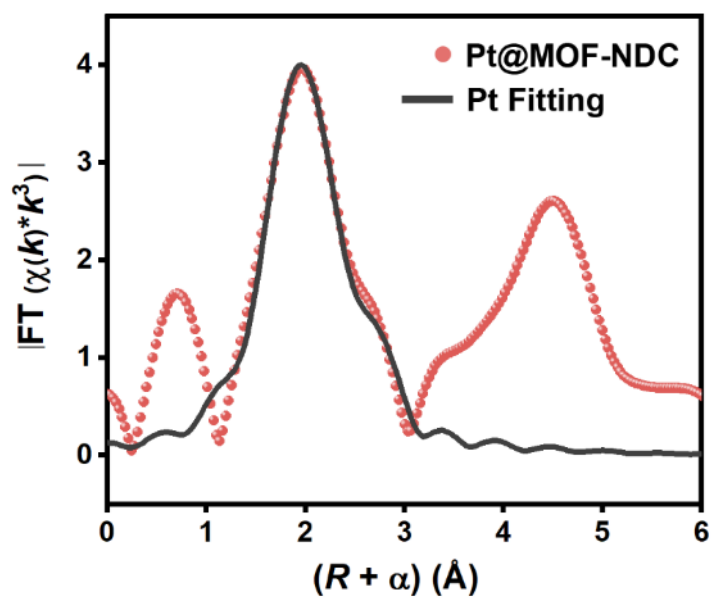


Figure S36. EXAFS fitting results of Pt for Pt@MOF-NDC.

Table S3. EXAFS fitting parameters at Pt L-edge for Pt@MOFs. ($S_0^2 = 0.826$)

Sample	Scattering pair	R (Å)	* σ^2 (Å ²)	ΔE_0 (eV)	R factor
Pt@MOF-BCP	Pt-N	1.96 ± 0.05	-	-0.9	0.0181
	Pt-Fe	2.52 ± 0.04	0.030		
	Pt-Pt	2.66 ± 0.01	0.011		
Pt@MOF-BDC	Pt-N	1.98 ± 0.07	0.013	-6.8 ± 3.7	0.0126
	Pt-Fe	2.63 ± 0.15	0.024		
	Pt-Pt	2.88 ± 0.24	0.021		
Pt@MOF-NDC	Pt-N	2.03 ± 0.11	-	-9.9	0.0173
	Pt-Fe	2.50 ± 0.02	0.023		
	Pt-Pt	2.64 ± 0.00	0.017		
Pt foil	Pt-Pt	2.76 ± 0.12	0.005	9.5 ± 0.7	0.0072

S_0^2 : Amplitude reduction factor; R (Å): Interatomic distance;

* σ^2 (Å²): Debye-Waller factor; In Pt@MOF-BCP and Pt@MOF-NDC, Debye-Waller: debye (temp, thetad). Here, it should be noted that the EXAFS analysis obtained is the average coordination result. Data ranges of Pt@MOF-BCP: $3.3 \leq k \leq 10.7 \text{ Å}^{-1}$, $1.1 \leq R \leq 3.3 \text{ Å}$. The number of variable parameters is 7, out of a total of 10.2 independent points. R-factor for this fit is 1.8 %.

Data ranges of Pt@MOF-BDC: $2.7 \leq k \leq 10.3 \text{ Å}^{-1}$, $1.0 \leq R \leq 3.0 \text{ Å}$. The number of variable parameters is 7, out of a total of 9.4 independent points. R-factor for this fit is 1.2 %.

Data ranges of Pt@MOF-NDC: $2.8 \leq k \leq 8.6 \text{ Å}^{-1}$, $1.2 \leq R \leq 3.1 \text{ Å}$. The number of variable parameters is 7, out of a total of 7.6 independent points. R-factor for this fit is 1.7 %.

Data ranges of Pt foil: $3.4 \leq k \leq 12.6 \text{ Å}^{-1}$, $1.5 \leq R \leq 3.1 \text{ Å}$. The number of variable parameters is 4, out of a total of 9.3 independent points. R-factor for this fit is 0.7 %.

Here, it should be noted that the EXAFS analysis obtained is the average coordination result.

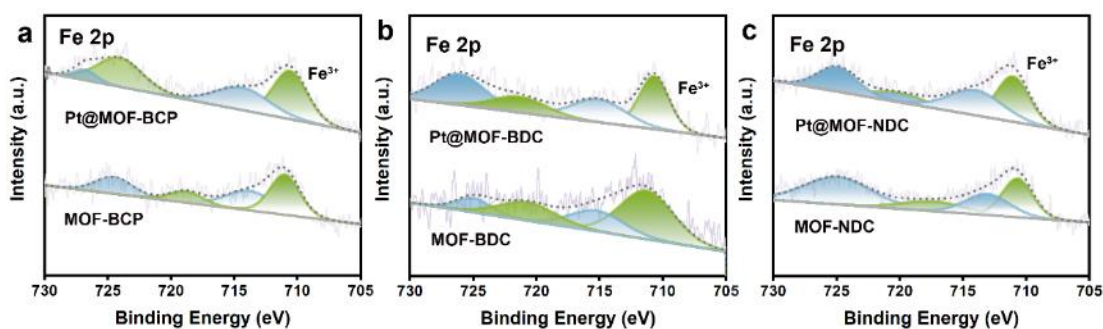


Figure S37. XPS spectra of Fe 2p for Pt@MOFs and the corresponding MOF precursors.

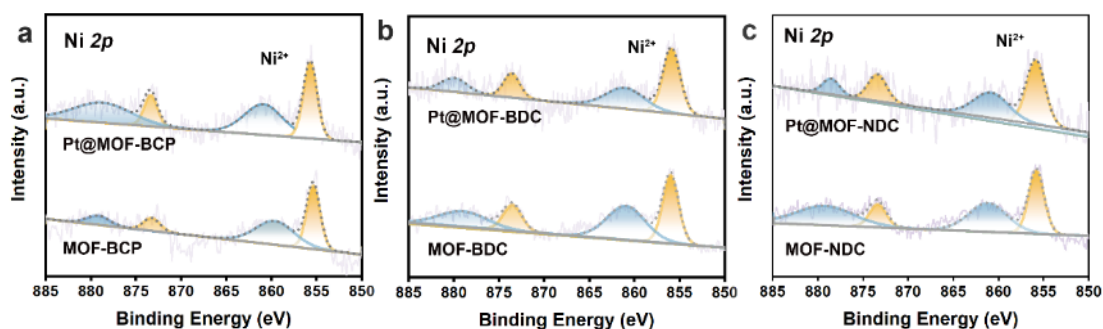


Figure S38. XPS spectra of Ni 2p for Pt@MOFs and the corresponding MOF precursors.

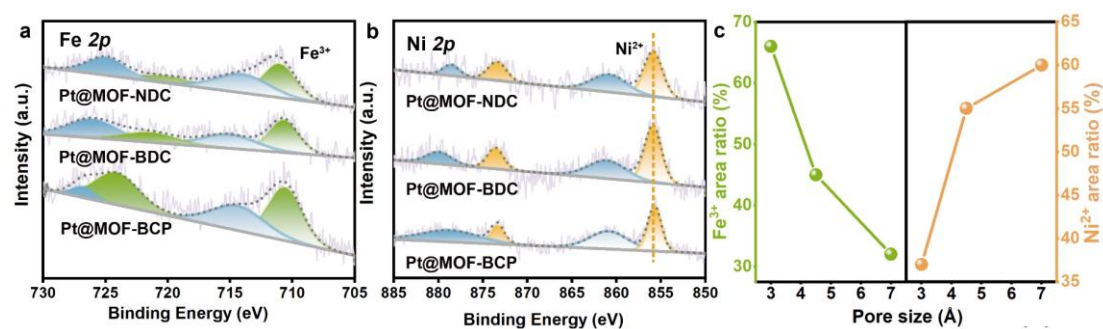


Figure S39. (a) XPS spectra of Fe 2p for Pt@MOFs. (b) XPS spectra of Ni 2p for Pt@MOFs. (c) The area ratio of Fe³⁺ and Ni²⁺ of Pt@MOFs.

Note: The gradual increase in Fe^{III} and the decrease in Ni^{II} area ratios with decreasing pore size of MOF precursor indicate the presence of more high-valent metal center in Pt@MOF-BCP, which can be attributed to the stronger Pt-support interaction.

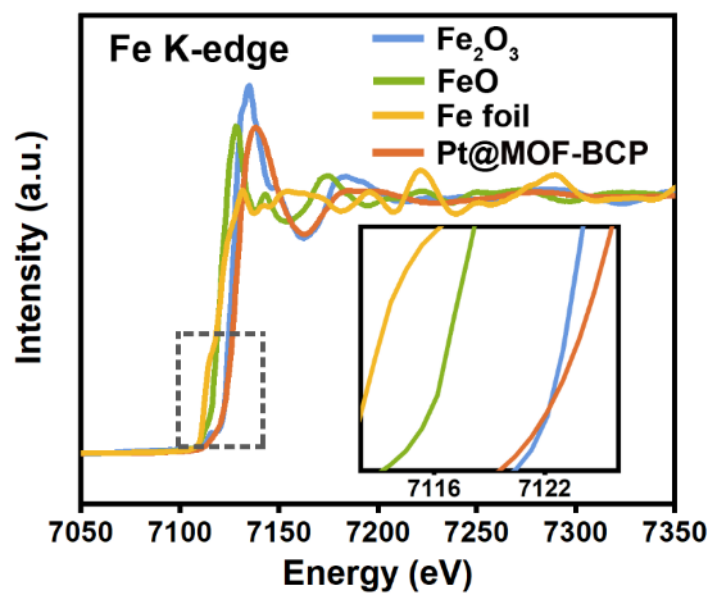


Figure S40. Normalized Fe K-edge XANES spectra for Pt@MOF-BCP, Fe₂O₃, FeO and Fe foil.

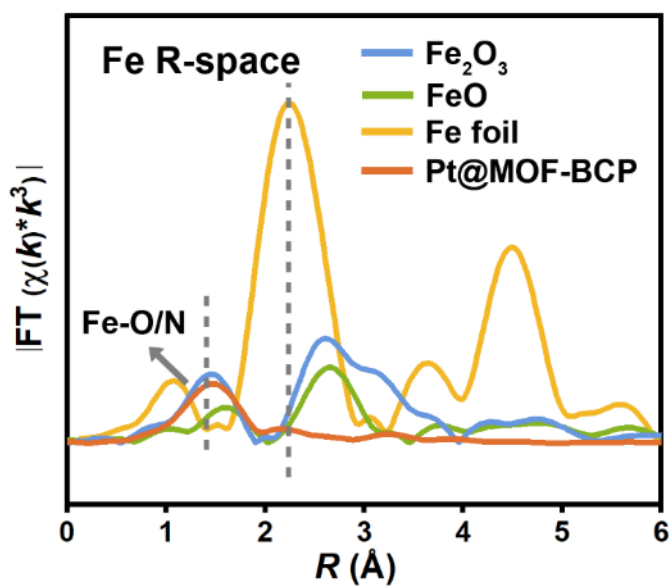


Figure S41. FT-EXAFS spectra of Fe K-edge in Pt@MOF-BCP.

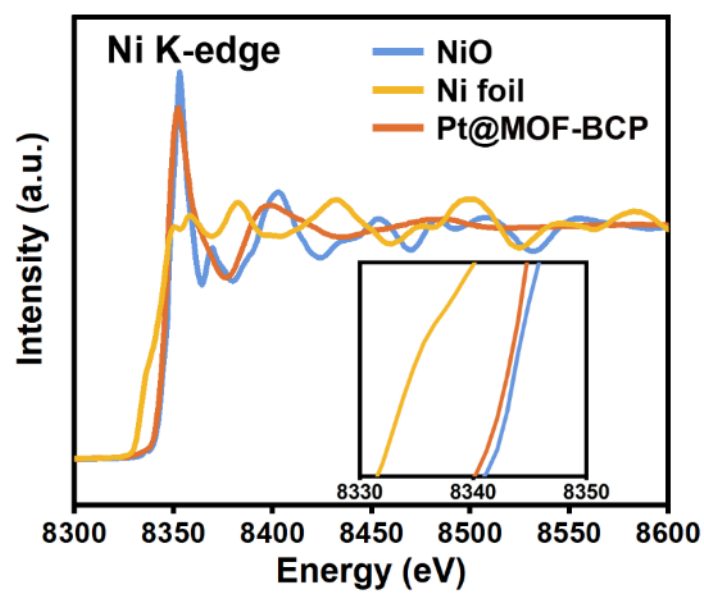


Figure S42. Normalized Ni K-edge XANES spectra for Pt@MOF-BCP, NiO and Ni foil.

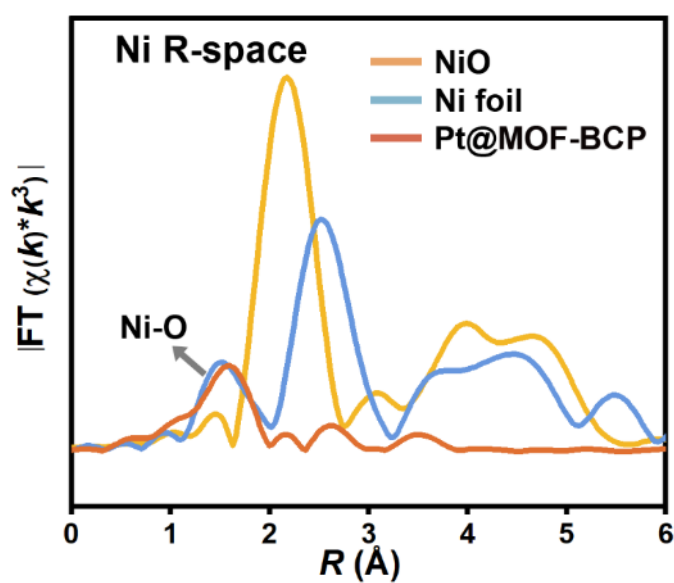


Figure S43. FT-EXAFS spectra of Ni K-edge in Pt@MOF-BCP.

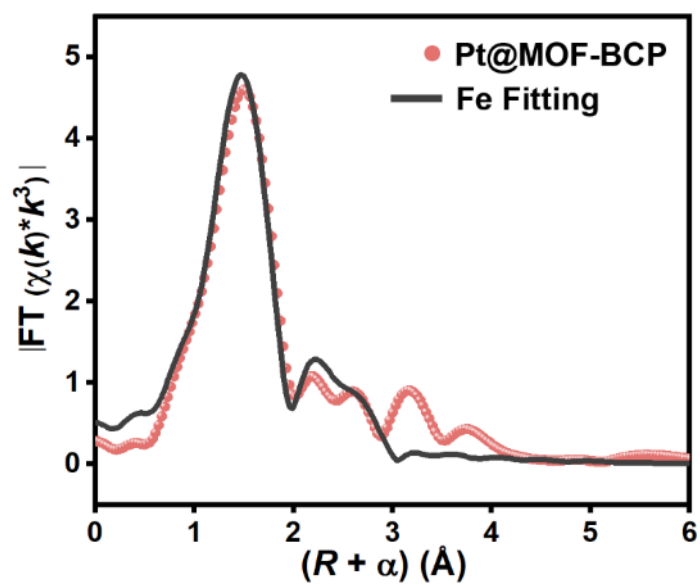


Figure S44. EXAFS fitting results of Fe for Pt@MOF-BCP.

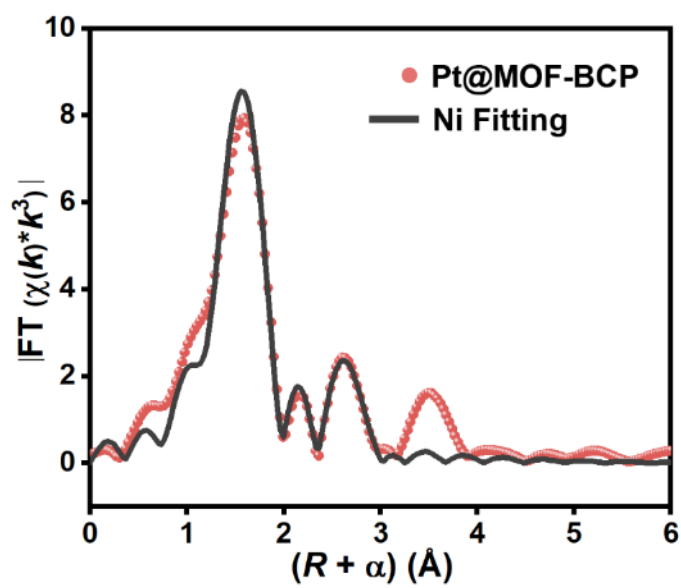


Figure S45. EXAFS fitting results of Ni for Pt@MOF-BCP.

Table S4. EXAFS fitting parameters at Fe K-edge for Pt@MOF-BCP sample. ($S_0^2 = 0.826$)

Sample	Scattering pair	R (Å)	* σ^2 (Å ²)	ΔE_0 (eV)	R factor
Pt@MOF-BCP	Fe-O	2.00 ± 0.17	0.013	-3.7 ± 3.1	0.0196
	Fe-Fe	2.64 ± 0.22	0.031		
	Fe-Pt	2.62 ± 0.02	0.016		

Data ranges: $2.5 \leq k \leq 9.5 \text{ Å}^{-1}$, $1.0 \leq R \leq 3.0 \text{ Å}$. The number of variable parameters is 7, out of a total of 8.7 independent points. R-factor for this fit is 1.9 %.

Table S5. EXAFS fitting parameters at Ni K-edge for Pt@MOF-BCP sample. ($S_0^2 = 0.826$)

Sample	Scattering pair	R (Å)	* σ^2 (Å ²)	ΔE_0 (eV)	R factor
Pt@MOF-BCP	Ni-O	2.05 ± 0.03	0.006	-3.9 ± 1.3	0.0133
	Ni-Ni	2.55 ± 0.11	0.034		
	Ni-Pt	2.67 ± 0.02	0.018		

Data ranges: $2.5 \leq k \leq 10.3 \text{ Å}^{-1}$, $1.0 \leq R \leq 3.0 \text{ Å}$. The number of variable parameters is 7, out of a total of 9.7 independent points. R-factor for this fit is 1.3 %.

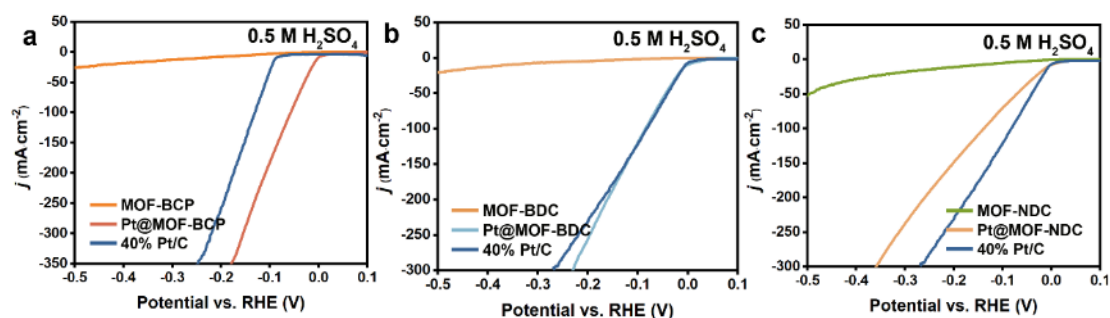


Figure S46. LSV curves of MOF precursors, Pt@MOFs and 40% Pt/C for HER in 0.5 M H₂SO₄.

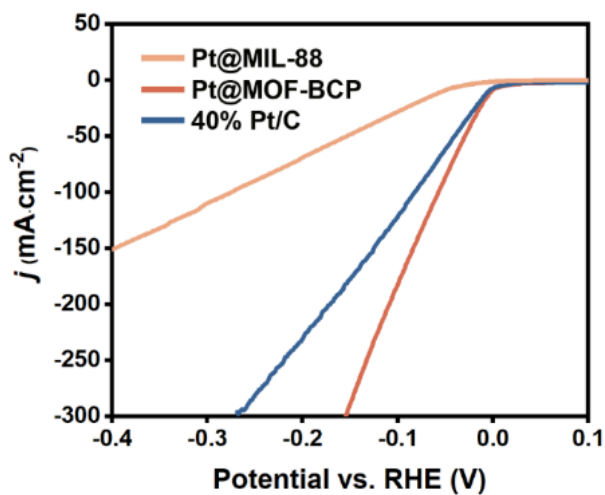


Figure S47. LSV curves of Pt@MIL-88, Pt@MOF-BCP and 40% Pt/C in 0.5 M H₂SO₄ for HER.

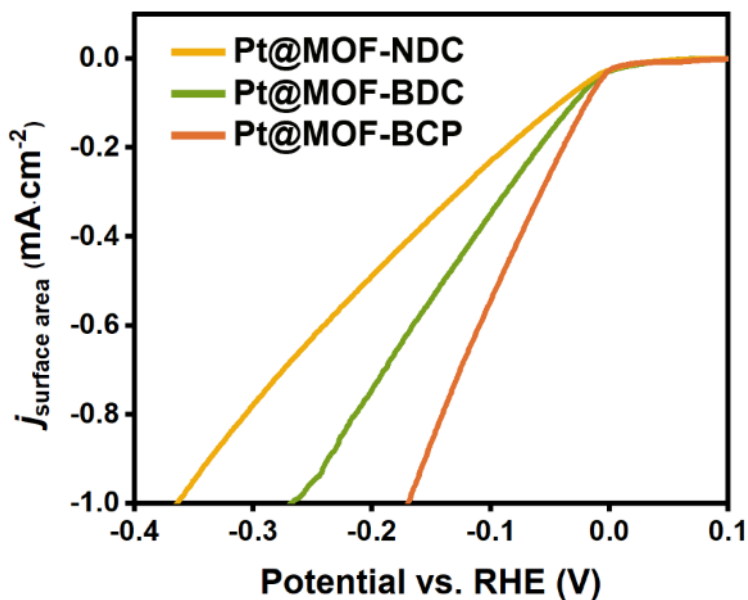


Figure S48. BET-normalized LSV curves of Pt@MOFs.

Note: BET-normalized LSV curves indicated that Pt@MOF-BCP required the lowest initial overpotential to drive the HER, indicating its superior intrinsic HER catalytic activity.

Table S6. The TOF value and mass activity of Pt@MOFs and Pt/C for HER in 0.5 M H₂SO₄ at $\eta = 100$ mV.

Sample	TOF (s ⁻¹)	Mass activity (A·g ⁻¹)
Pt@MOF-BCP	0.249	122.1
Pt@MOF-BDC	0.173	80.3
Pt@MOF-NDC	0.107	47.1
Pt/C	0.211	81.6

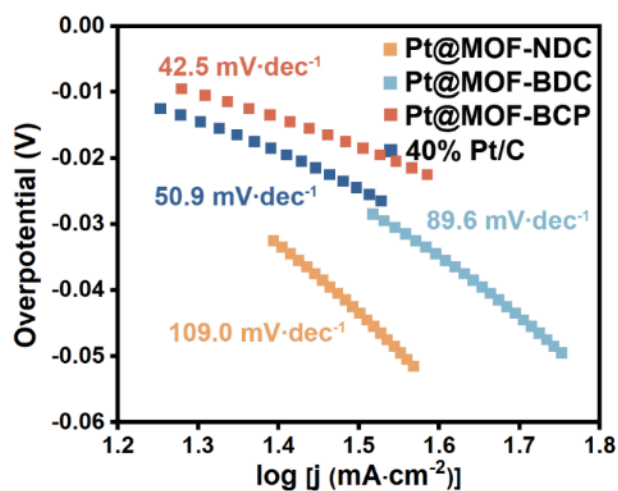


Figure S49. Tafel slopes of Pt@MOFs and 40% Pt/C for HER in 0.5 M H₂SO₄.

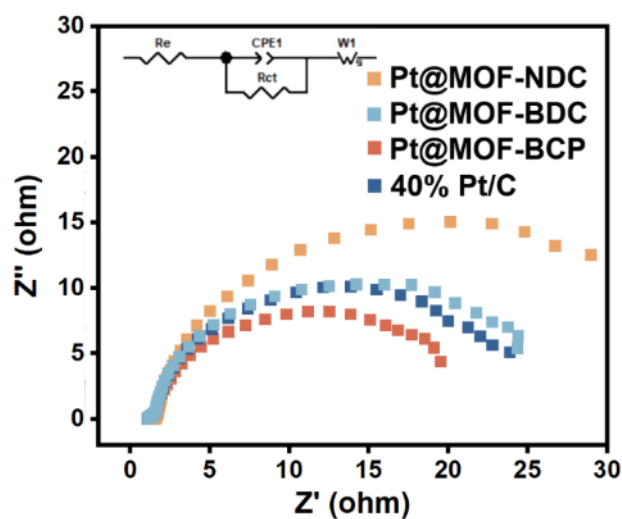


Figure S50. The Electrochemical impedance spectra (EIS) of Pt@MOFs and 40% Pt/C for HER in 0.5 M H₂SO₄.

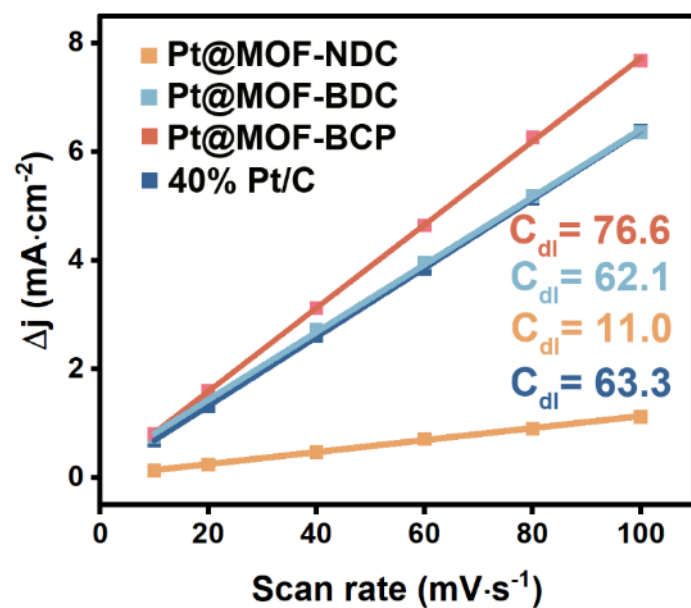


Figure S51. C_{dl} calculation of Pt@MOFs and 40% Pt/C for HER in 0.5 M H_2SO_4 .

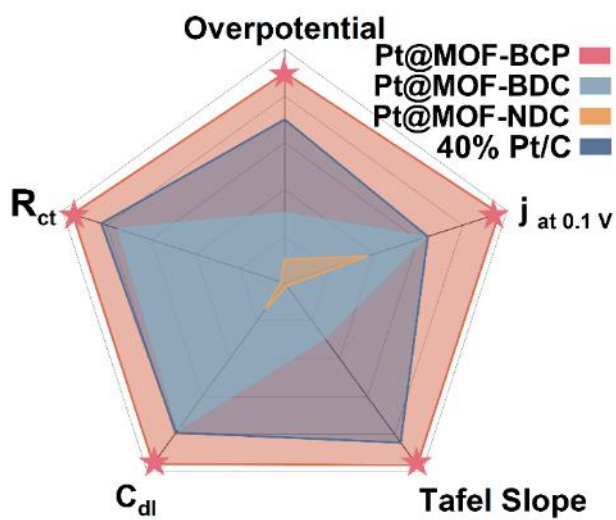


Figure S52. Comparison of performance metrics of Pt@MOFs and 40% Pt/C for HER.

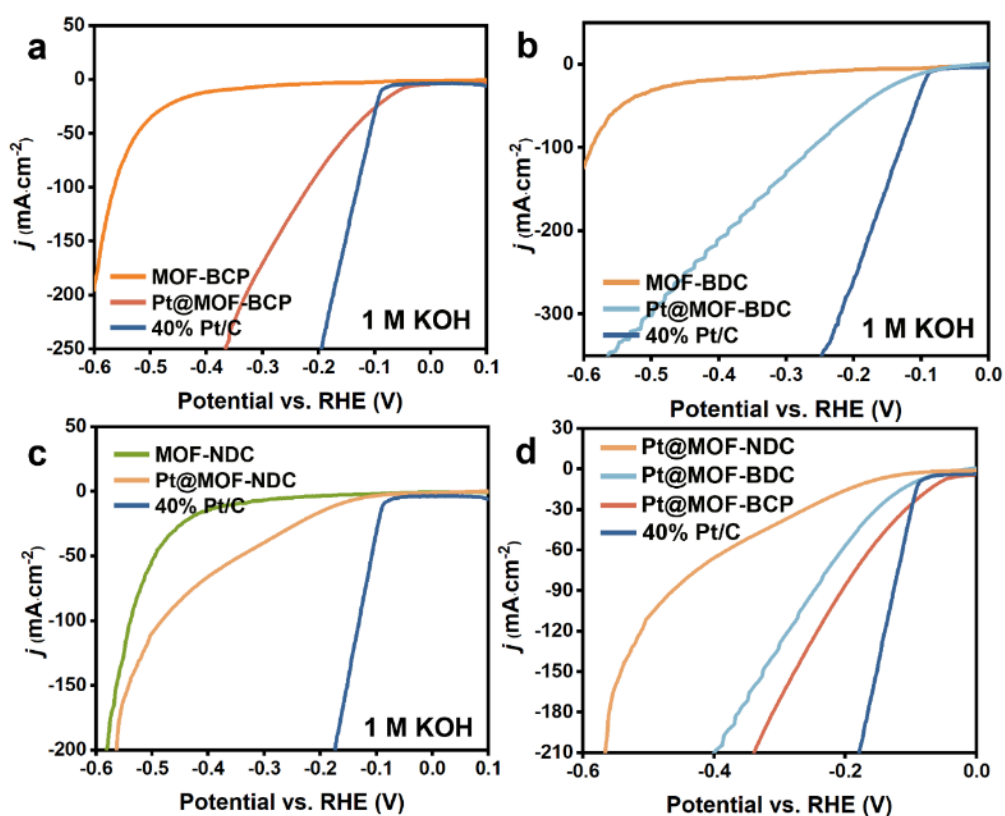


Figure S53. (a-c) LSV curves of MOF precursors, Pt@MOFs and 40% Pt/C for HER in 1 M KOH. (d) LSV curves of Pt@MOFs and 40% Pt/C for HER in 1 M KOH.

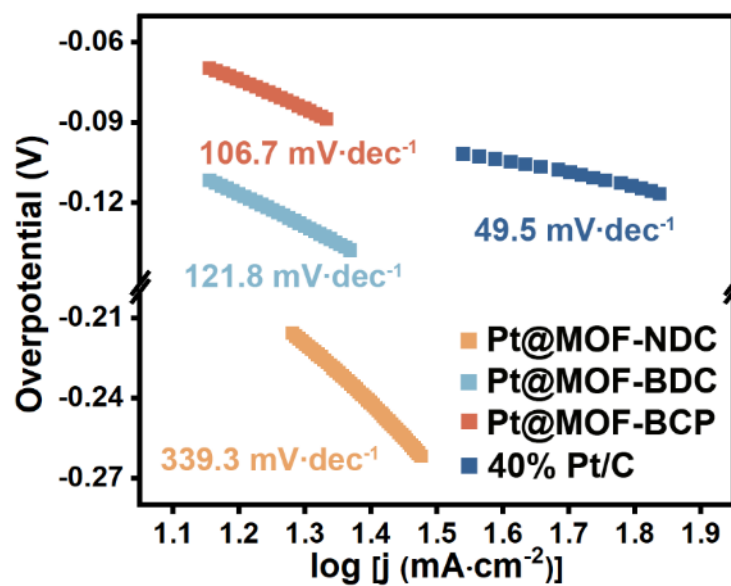


Figure S54. Tafel slopes of Pt@MOFs and 40% Pt/C for HER in 1 M KOH.

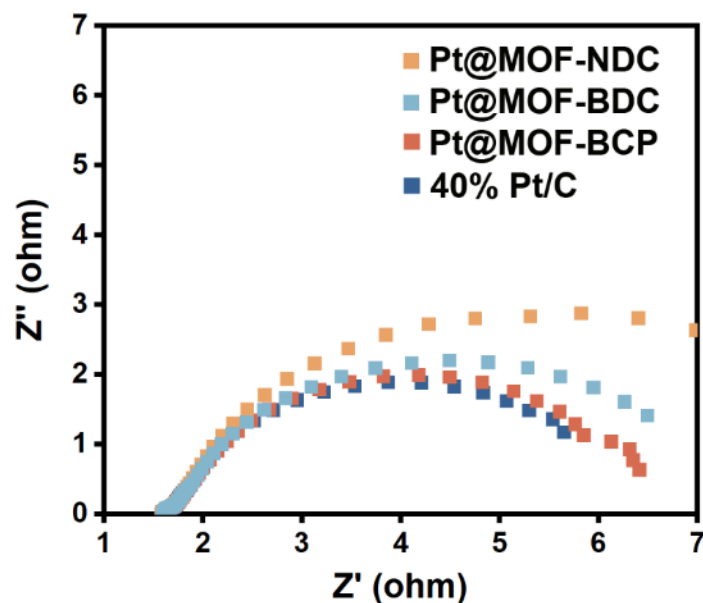


Figure S55. The Electrochemical impedance spectra (EIS) of Pt@MOFs and 40% Pt/C for HER in 1 M KOH.

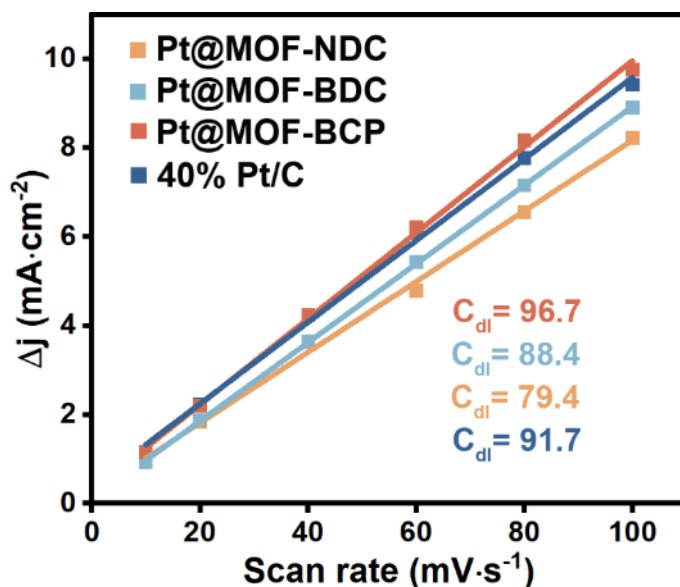


Figure S56. C_{dl} calculation of Pt@MOFs and 40% Pt/C for HER in 1 M KOH.

Note: Impressively, a current density of $10 \text{ mA} \cdot \text{cm}^{-2}$ for Pt@MOF-BCP was obtained at an ultralow overpotential of 56.7 mV in 1 M KOH for HER, highlighting its promising applicability across a broad pH range (Figure S53). Tafel value of Pt@MOF-BCP ($106.7 \text{ mV} \cdot \text{dec}^{-1}$) was smaller than that of other sample, demonstrating faster HER kinetics in alkaline media (Figure S54). R_{ct} of Pt@MOF-BCP was low than that of Pt@MOF-BDC and Pt@MOF-NDC catalysts, primarily due to the excellent mass transfer by modulated electronic structure of Pt (Figure S55). Additionally, the larger

C_{dl} value and higher specific activity further confirmed the superior intrinsic HER activity of Pt@MOF-BCP in 1 M KOH (Figure S56).

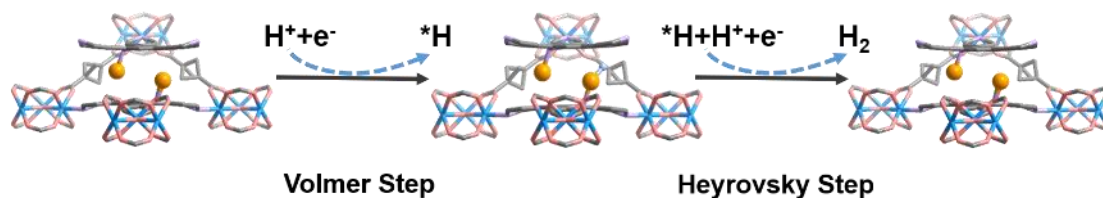


Figure S57. Modeling of the elementary electrochemical steps for HER on the Pt site of Pt@MOF-BCP.

Note: The first step involves the Volmer reaction, where protons are reduced to form adsorbed hydrogen on the Pt site of the catalyst ($H^+ + e^- \rightarrow H^*$). Subsequently, the followed by the Heyrovsky step, in which adsorbed H^* reacts with another proton and electron to generate a hydrogen molecule ($H^* + H^+ + e^- \rightarrow H_2$) (Figure S57).

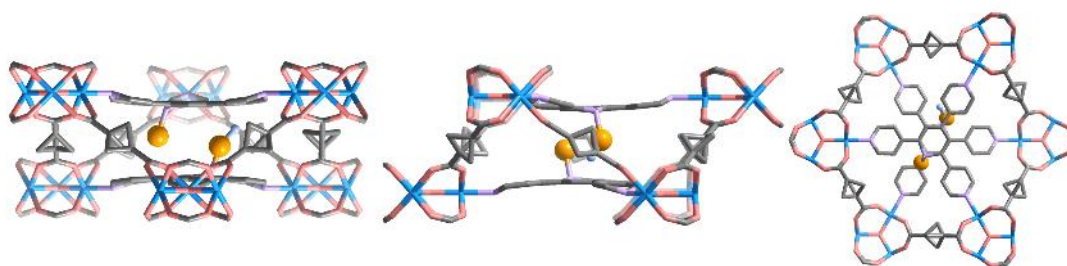


Figure S58. Schematic illustration of the crystal structure models of the Pt@MOF-BCP with adsorbed H^* intermediates.

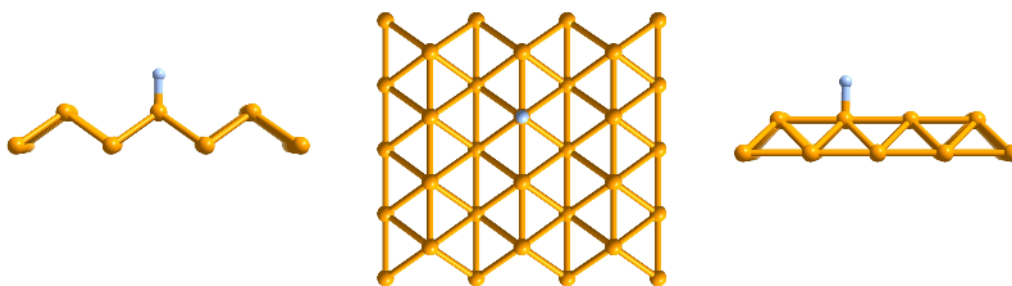


Figure S59. Schematic illustration of the crystal structure models of the Pt cluster with adsorbed H^* intermediates.

Note: To investigate the effect of noble metal loading, a series of $\text{Pt}_x\text{@MOF-BCP}$ catalysts with varying Pt contents were prepared, where x represents the input amount of Pt. SEM, TEM and HAADF-STEM images of $\text{Pt}_5\text{@MOF-BCP}$ and $\text{Pt}_{15}\text{@MOF-BCP}$ revealed that the spindle-like morphology was maintained after introducing Pt clusters (Figures S60-S65). And mapping images showed that the uniform distribution of Pt, Fe and Ni (Figures S66-S67). Meanwhile, $\text{Pt}_{15}\text{@MOF-BCP}$ showed the higher Pt loading content compared with $\text{Pt}_5\text{@MOF-BCP}$ (Figures S68-S69). Electrochemical analysis indicated that the HER performance in 0.5 M H_2SO_4 and 1 M KOH of $\text{Pt}_{10}\text{@MOF-BCP}$ (named as Pt@MOF-BCP in the manuscript) (57.5 mV/218.7 mV) was much better than that of $\text{Pt}_5\text{@MOF-BCP}$ (110.5 mV/292.7 mV) and $\text{Pt}_{15}\text{@MOF-BCP}$ (131.5 mV/437.7 mV) at $100\text{ mA}\cdot\text{cm}^{-2}$ (Figure S70).

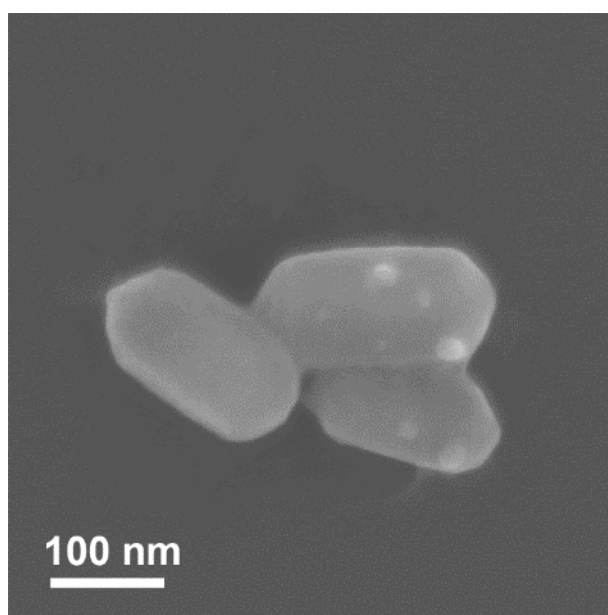


Figure S60. SEM image of $\text{Pt}_5\text{@MOF-BCP}$.

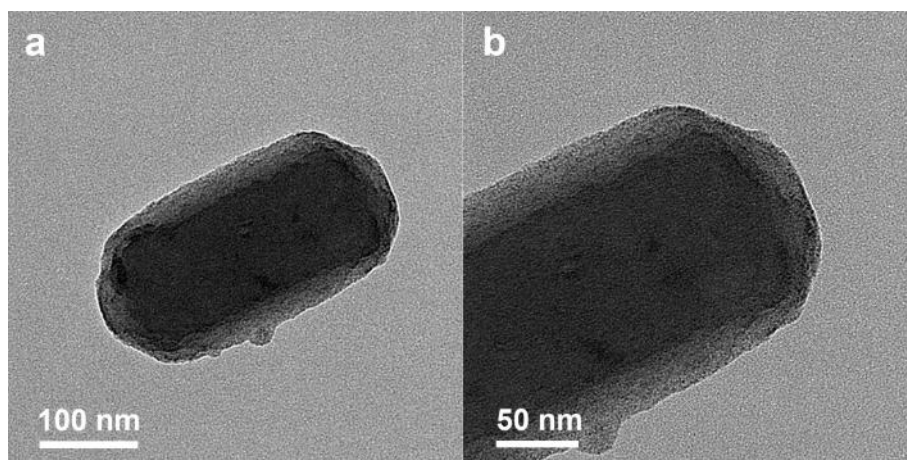


Figure S61. TEM images of $\text{Pt}_5\text{@MOF-BCP}$.

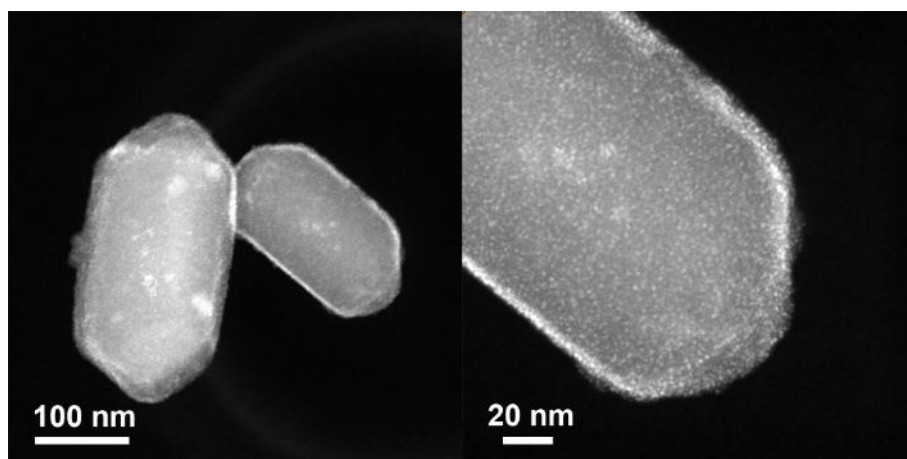


Figure S62. HAADF-STEM images of Pt₅@MOF-BCP.

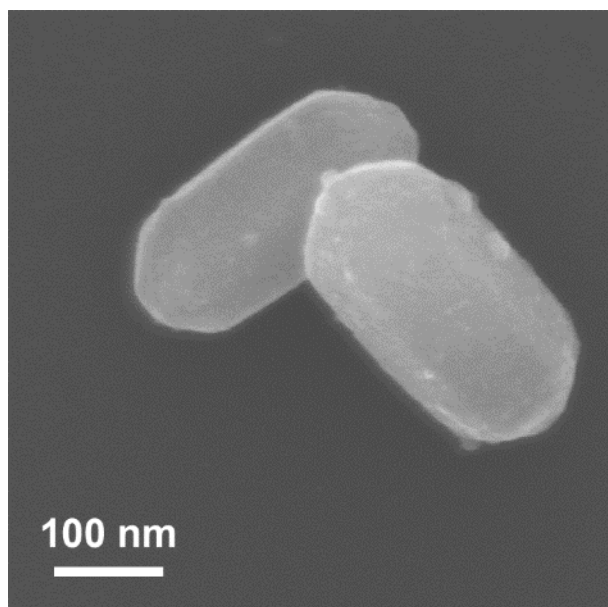


Figure S63. SEM image of Pt₁₅@MOF-BCP.

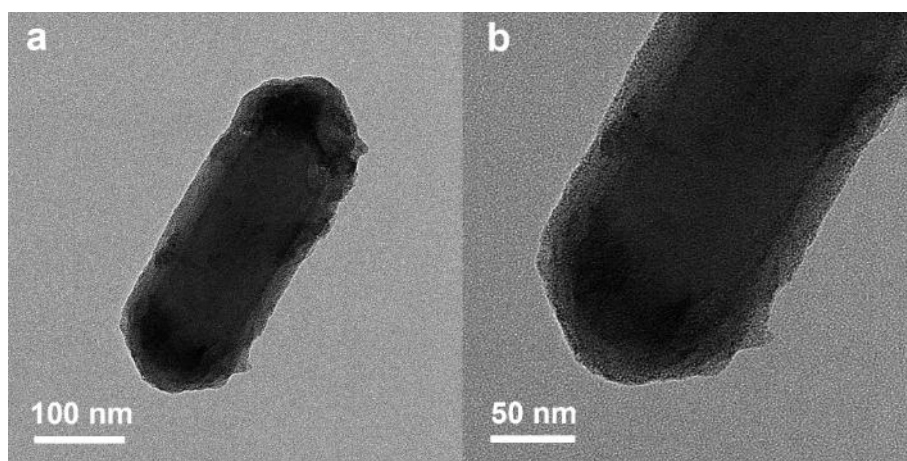


Figure S64. TEM images of Pt₁₅@MOF-BCP.

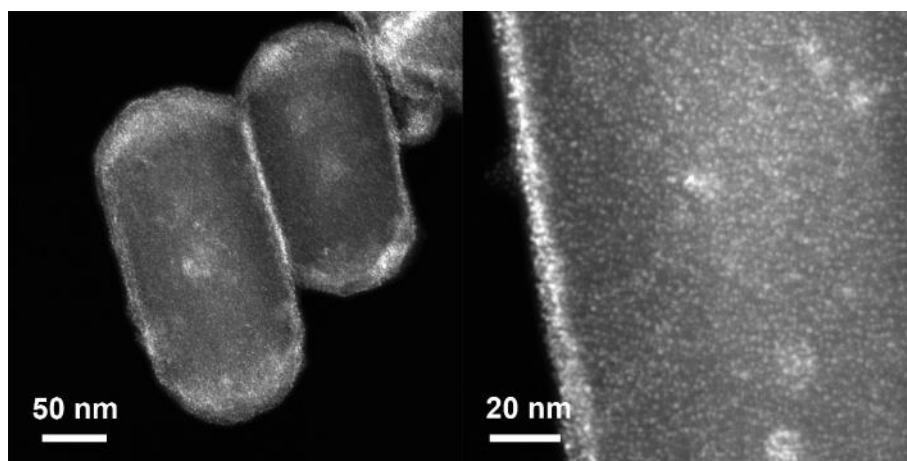


Figure S65. HAADF-STEM images of Pt₁₅@MOF-BCP.

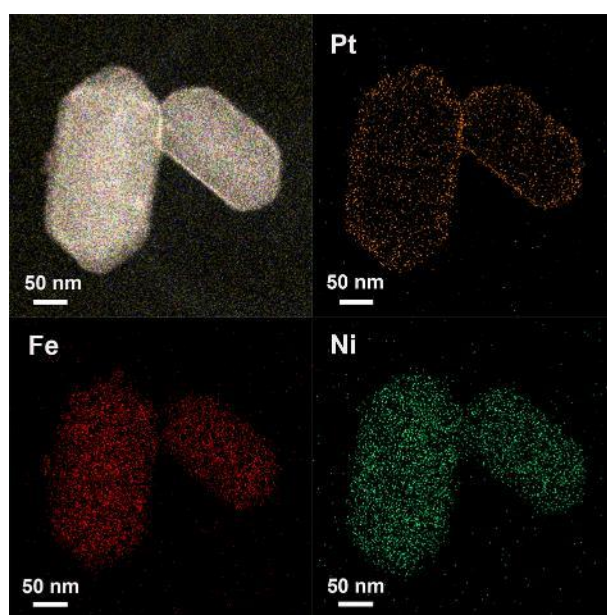


Figure S66. Mapping images of Pt₅@MOF-BCP.

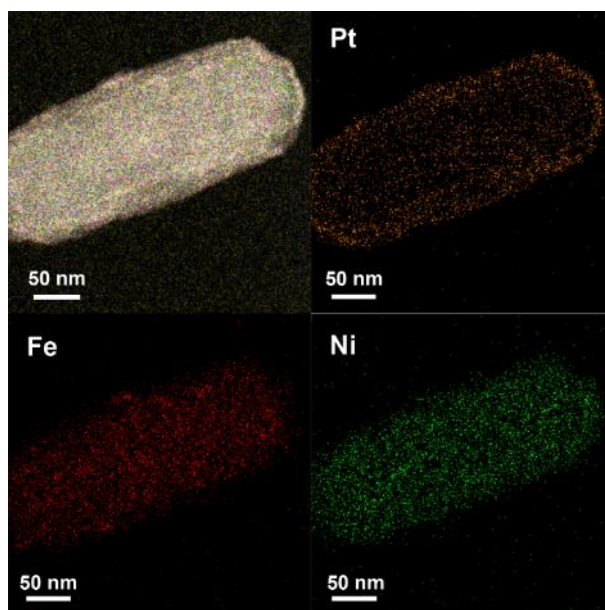


Figure S67. Mapping images of Pt₁₅@MOF-BCP.

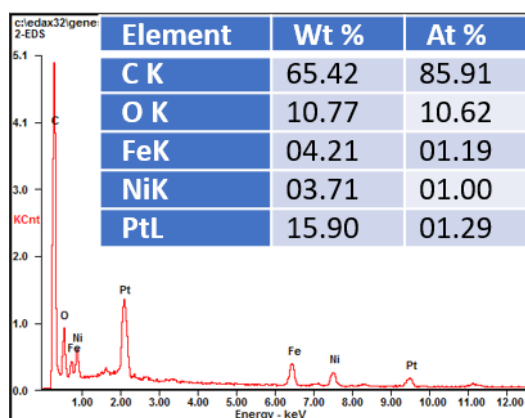


Figure S68. The EDX of Pt₅@MOF-BCP.

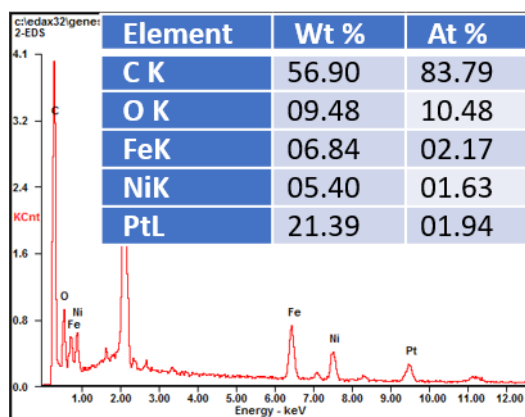


Figure S69. The EDX of Pt₁₅@MOF-BCP.

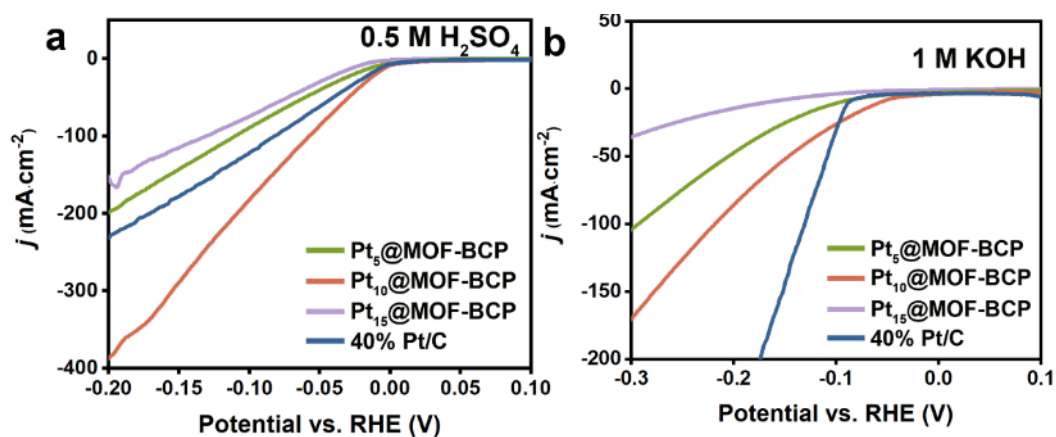


Figure S70. (a) LSV curves of Pt_x@MOF-BCP and 40% Pt/C for HER in 0.5 M H₂SO₄. (b) LSV curves of Pt_x@MOF-BCP and 40% PtC for HER in 1 M KOH. (Pt₁₀@MOF-BCP is abbreviated to Pt@MOF-BCP in manuscript).

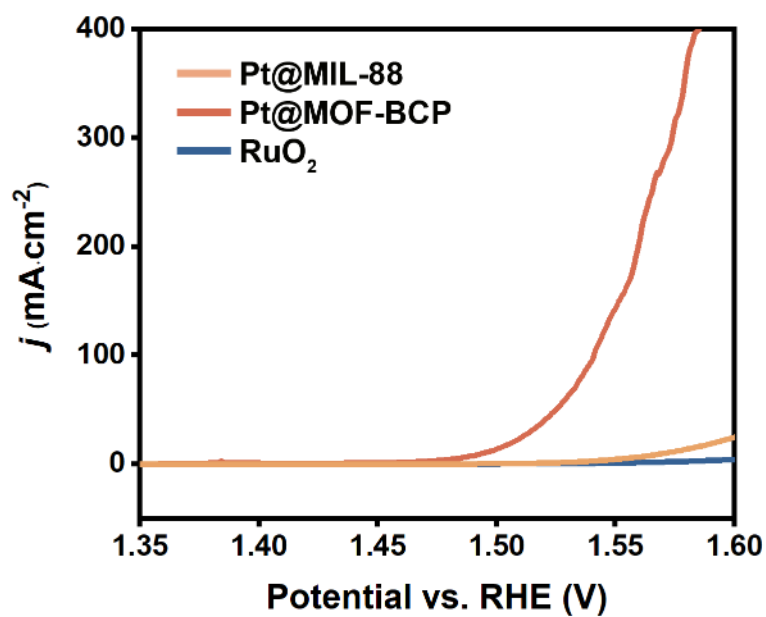


Figure S71. LSV curves of Pt@MIL-88, Pt@MOF-BCP and RuO₂ for OER in 1 M KOH.

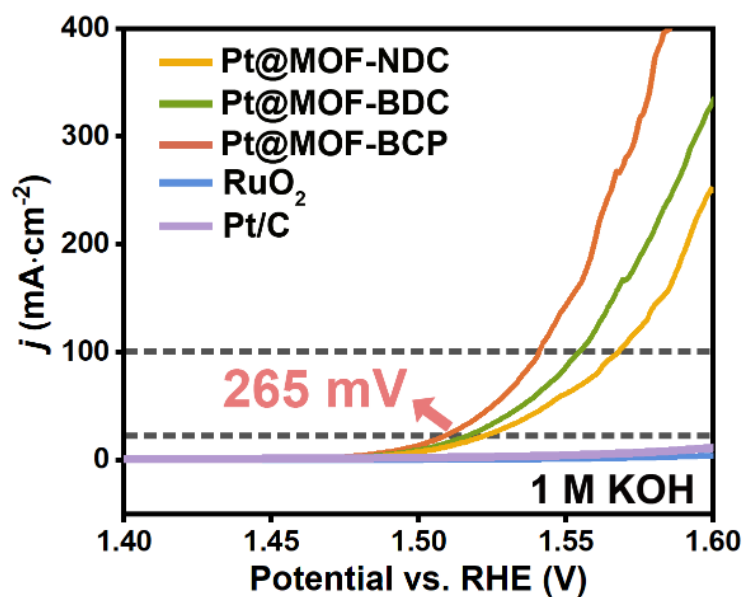


Figure S72. LSV curves of Pt@MOFs, Pt/C and RuO₂ for OER in 1 M KOH.

Table S7. The TOF value and mass activity of Pt@MOF-BCP, Pt/C and RuO₂ for OER in 1 M KOH at $\eta = 300$ mV.

Sample	TOF (s ⁻¹)	Mass activity (A·g ⁻¹)
Pt@MOF-BCP	0.043	42.1
RuO ₂	0.00002	0.406
Pt/C	0.00002	2.0

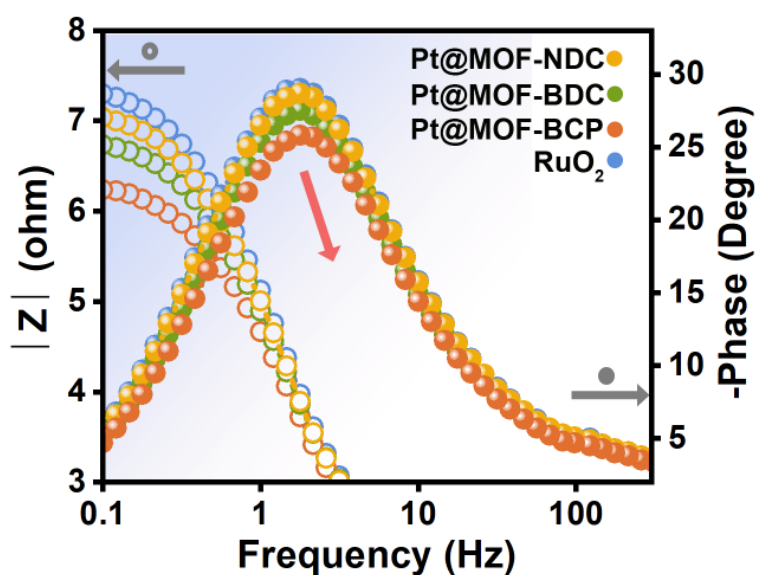


Figure S73. Bode plots of Pt@MOFs and RuO₂ for OER in 1 M KOH.

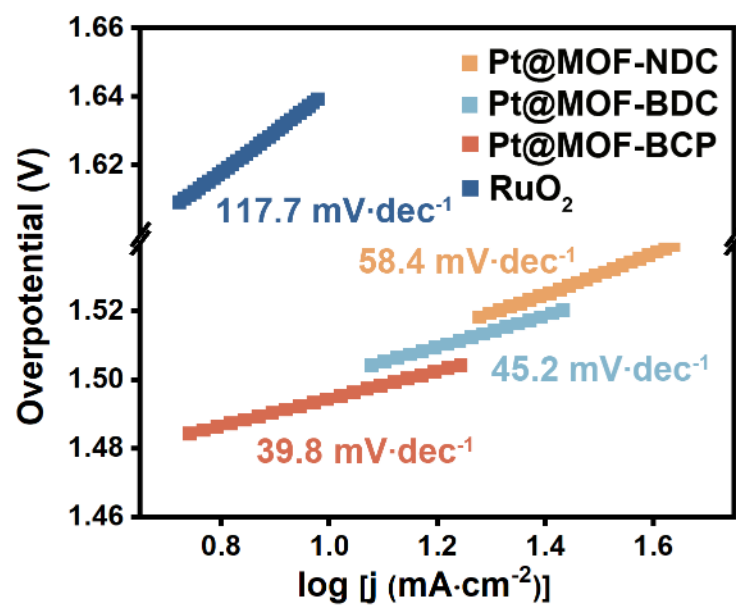


Figure S74. Tafel slopes of Pt@MOFs and RuO₂ for OER in 1 M KOH.

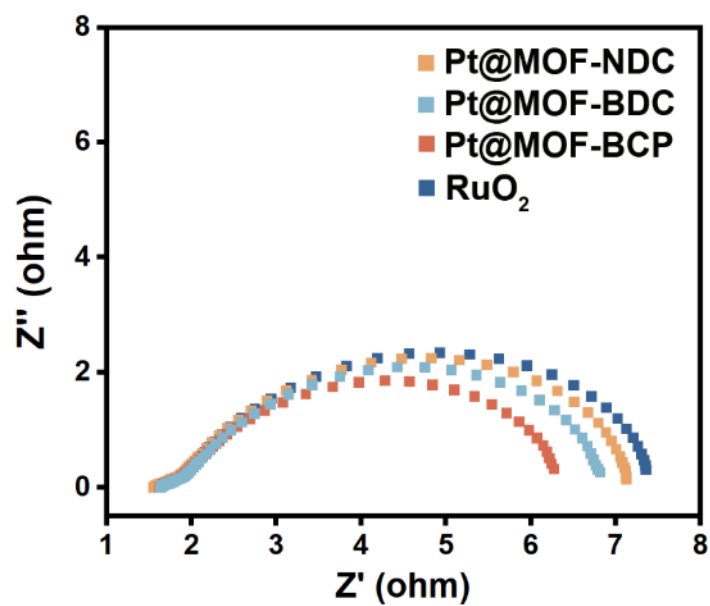


Figure S75. The Electrochemical impedance spectra (EIS) of Pt@MOFs and RuO₂ for OER in 1 M KOH.

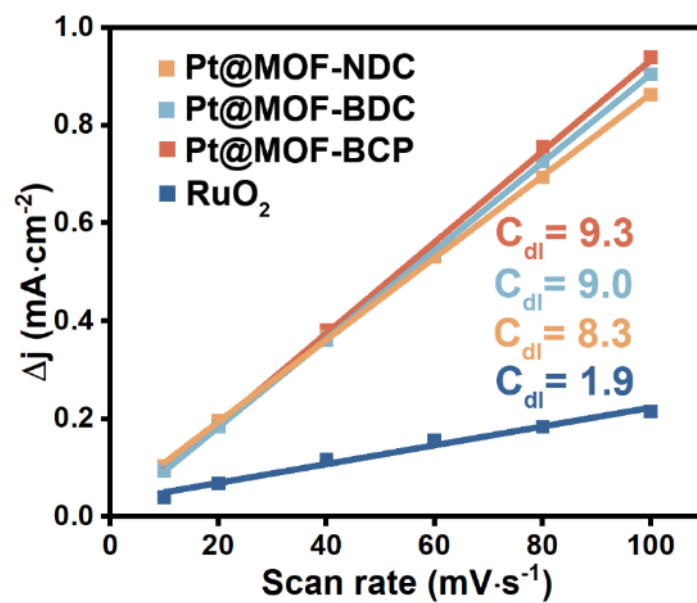


Figure S76. C_{dl} calculation of Pt@MOFs and RuO_2 for OER in 1 M KOH.

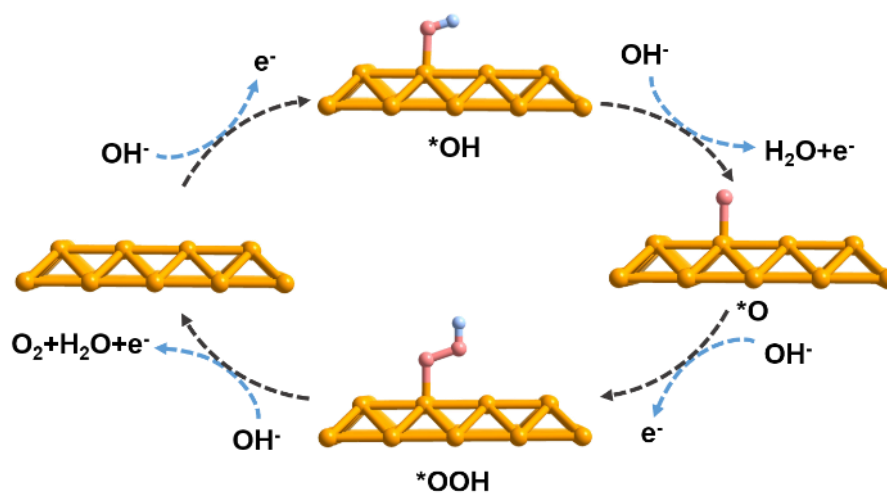


Figure S77. Modeling of the elementary electrochemical steps for OER on the Pt site of Pt cluster.

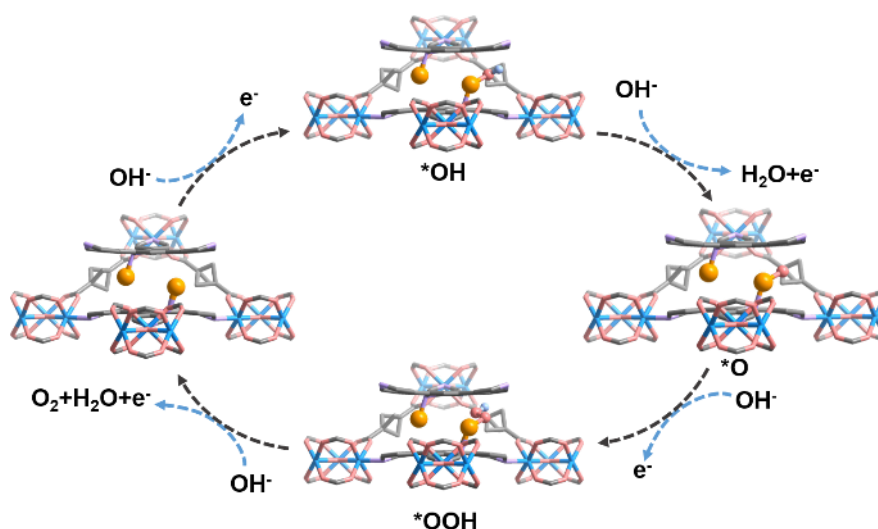


Figure S78. Modeling of the elementary electrochemical steps for OER on the Pt site of Pt@MOF-BCP.

Note: Following the four-step proton-coupled electron transfer (PCET) mechanism, the OER process in an alkaline solution was illustrated by local structural configurations with adsorbed intermediates (Figures S77-S78). Initially, OH^- is adsorbed onto the metal active site (M) to form M-OH^* , followed by deprotonation and electron transfer to generate M-O^* . Subsequently, the M-O^* undergoes a nucleophilic attack by another OH^- to form M-OOH^* intermediate. Finally, O_2 is released via deportation of the M-OOH^* , regenerating the metal active site for the next OER cycle.

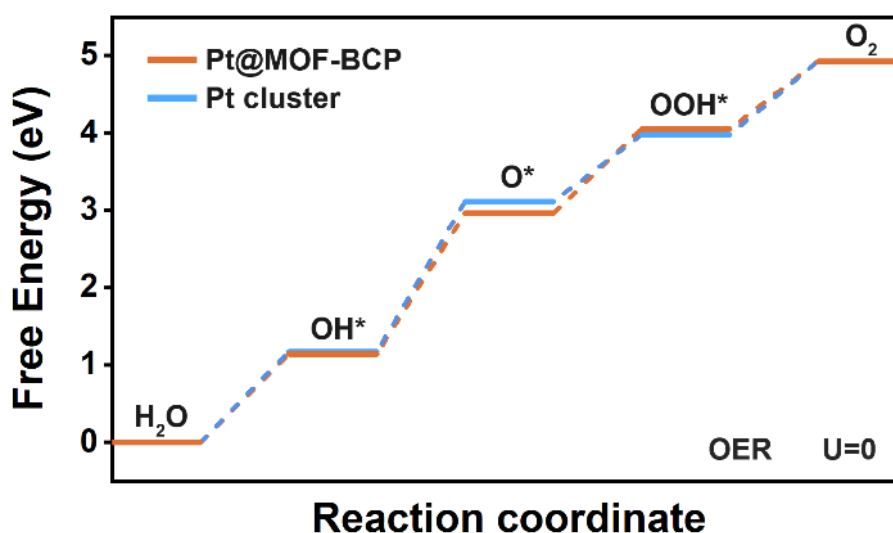


Figure S79. Free energy diagrams of the OER process on the Pt sites of Pt@MOF-BCP and Pt cluster at 0 V.

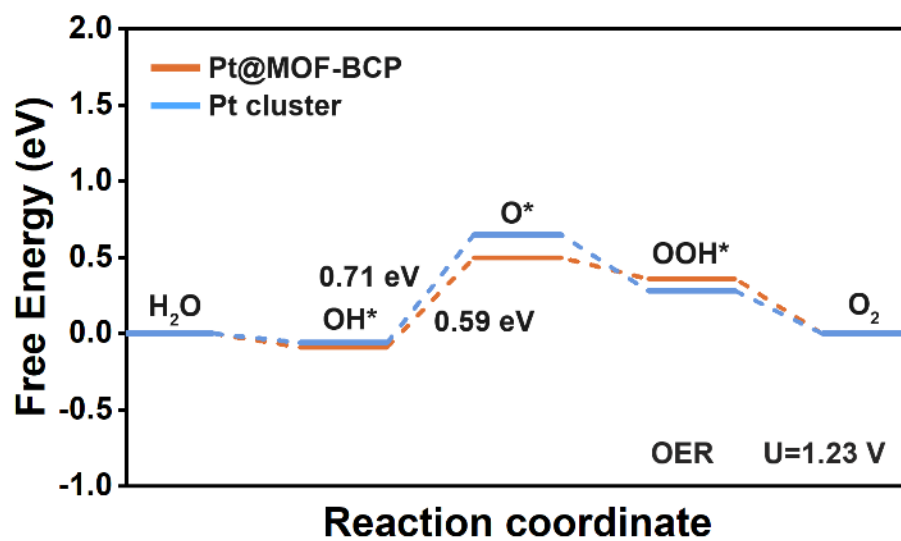


Figure S80. Free energy diagrams of the OER process on the Pt sites of Pt@MOF-BCP and Pt cluster at 1.23 V.

Table S8. Comparison of electrocatalytic performance for HER catalysts in 0.5 M H₂SO₄ and OER in 1 M KOH.

Catalyst	HER/OER Catalysis	η (mV) 10 mA·cm ⁻²	Ref.
Pt@MOF-BCP	HER (0.5 M H ₂ SO ₄)	2.5	This work
Pt@MOF-BDC	HER (0.5 M H ₂ SO ₄)	3.5	
Pt@MOF-NDC	HER (0.5 M H ₂ SO ₄)	6.5	
CoP/Co-MOF	HER (0.5 M H ₂ SO ₄)	27	Angew. Chem. Int. Ed. 2019, 58, 4679.
AB&CTGU-5 (1:4)	HER (0.5 M H ₂ SO ₄)	18	Angew. Chem. Int. Ed., 2017, 56, 13001.
CoBDC/MXene	HER (0.5 M H ₂ SO ₄)	29	Adv. Funct. Mater., 2023, 33, 2210322.
Pt/MOF-O	HER (0.5 M H ₂ SO ₄)	29	J. Am. Chem. Soc. 2021, 143, 16512.
P-Ni MOF/POM	HER (0.5 M H ₂ SO ₄)	35	J. Mater. Chem. A, 2023, 11, 15663-15669.
Co-BPDC	HER (0.5 M H ₂ SO ₄)	37	Small, 2023, 19, 2301850.
Ir@Ni-NDC	HER (0.5 M H ₂ SO ₄)	41	Angew. Chem. Int. Ed. 2023, 62, 202302220.
PtCu-MoO ₂ @C	HER (0.5 M H ₂ SO ₄)	42	J. Mater. Chem. A, 2020, 8, 19348.
MOF-808-EDTA	HER (0.5 M H ₂ SO ₄)	42.1	J. Mater. Chem. A, 2019, 7, 20239.
Pt@UiO-66-SH	HER (0.5 M H ₂ SO ₄)	57	Inorg. Chem. 2022, 61, 13271.
MIL-(IrNiFe)@NF	HER (0.5 M H ₂ SO ₄)	69	J. Mater. Chem. A, 2023, 11, 15663.
NU-1000	HER (0.5 M H ₂ SO ₄)	101	Inorg. Chem., 2023, 62, 7195.
Pt@MOF-BCP	OER (1 M KOH)	265	This work
Pt@MOF-BDC	OER (1 M KOH)	271	
Pt@MOF-NDC	OER (1 M KOH)	275	
Cu-doped NiFe MOFs	OER (1 M KOH)	200	Adv. Energy Mater., 2023, 13, 2301224.
2D MOF NSs	OER (1 M KOH)	221	Small, 2022, 18, 2201076.
Ni-M@C-400	OER (1 M KOH)	229	Nano Res., 2022, 15, 5884.
NiFe-btz	OER (1 M KOH)	239	Adv. Sci., 2022, 9, 2203712.
FeNiBDC/SSM	OER (1 M KOH)	239	Appl. Catal. B Environ., 2023, 325, 122343.
CTGU-10a2-d2	OER (1 M KOH)	240	Angew. Chem. Int. Ed., 2019, 58, 4227.
Ni ₈ Co ₂ -BDC	OER (1 M KOH)	274	Adv. Funct. Mater., 2023, 34, 2304380.
FN-2	OER (1 M KOH)	275	Small, 2019, 15, 1903410.
Fe(py) ₂ Ni(CN) ₄	OER (1 M KOH)	285	Inorg. Chem., 2022, 61, 7095.

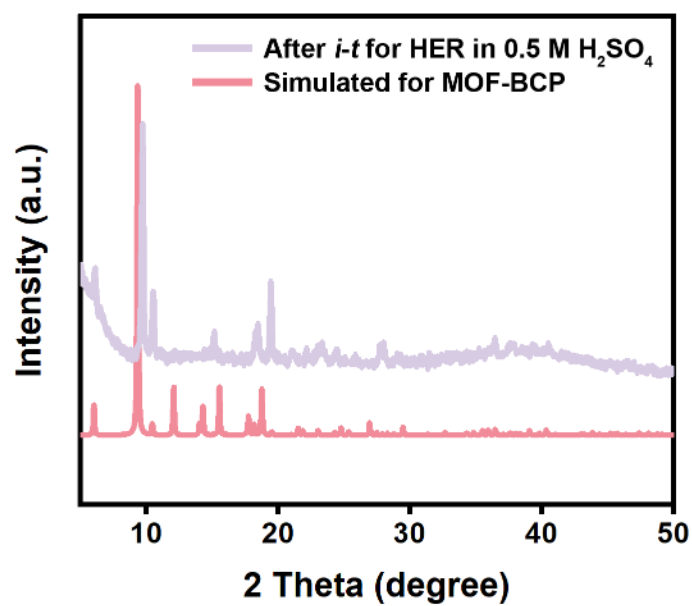


Figure S81. XRD pattern of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

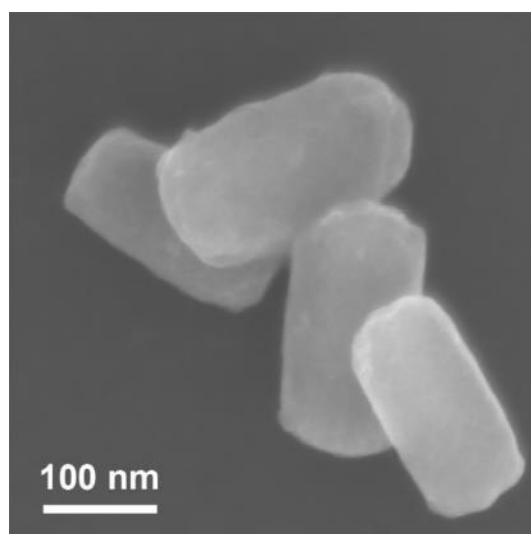


Figure S82. SEM image of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

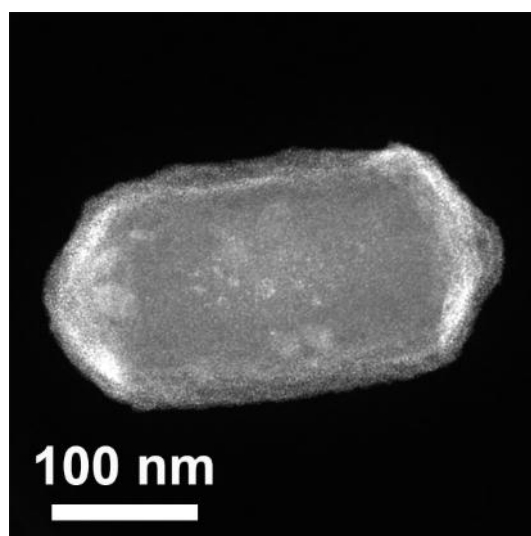


Figure S83. HAADF-STEM image of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

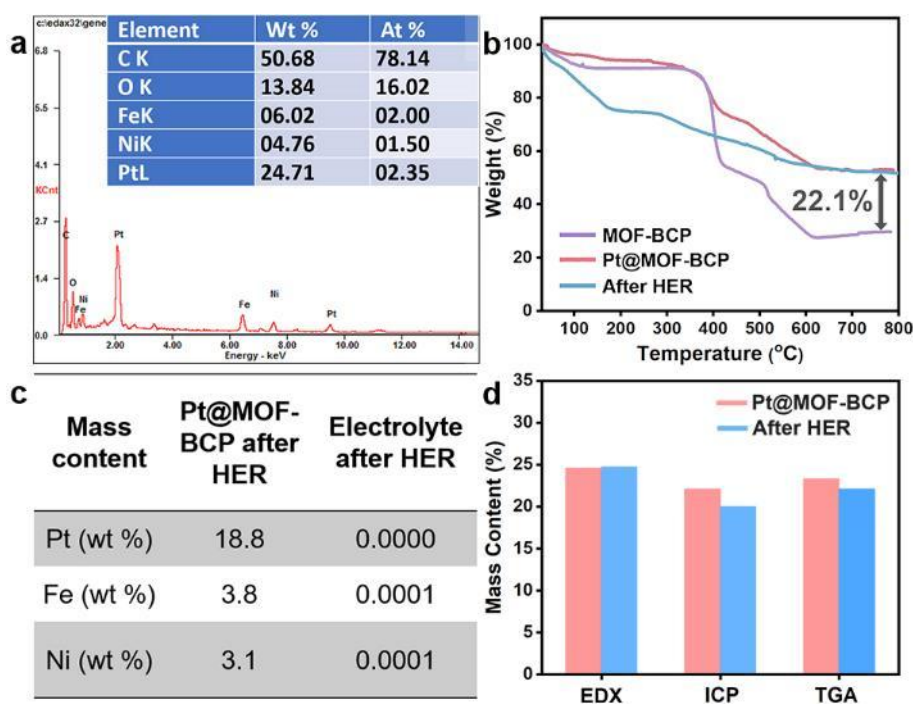


Figure S84. (a) The EDX of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄. (b) The thermogravimetric analyses (TGA) curves of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄. (c) The content of Pt, Fe and Ni of Pt@MOF-BCP in 40 mL 0.5 M H₂SO₄ electrolyte after HER test. (d) Comparison of the Pt mass content after HER test with the pristine sample in 0.5 M H₂SO₄.

Note: The post-reaction electrolyte was collected and analyzed after 1000 cyclic voltammetry (CV) cycles using a carbon cloth-supported catalyst with a loading of 1.5 mg·cm⁻², enabling the evaluation of its electrochemical durability under operational conditions. The post-reaction electrolyte was then diluted tenfold to analyze concentrations of metallic species *via* ICP-OES.

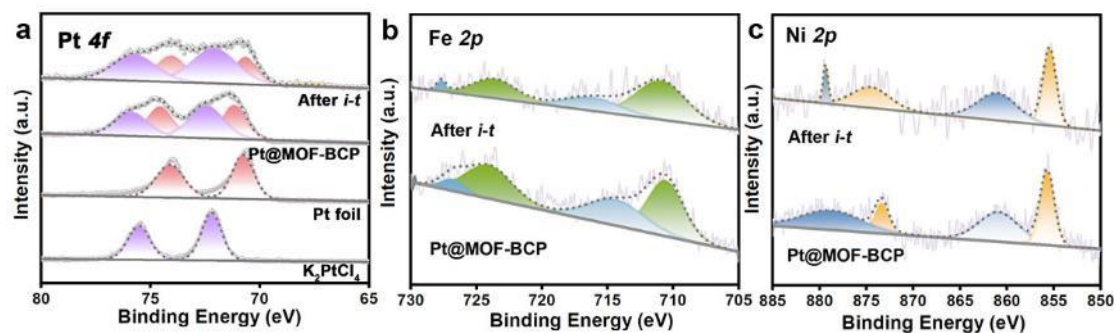


Figure S85. XPS spectra of (a) Pt 4*f*, (b) Fe 2*p* and (c) Ni 2*p* of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

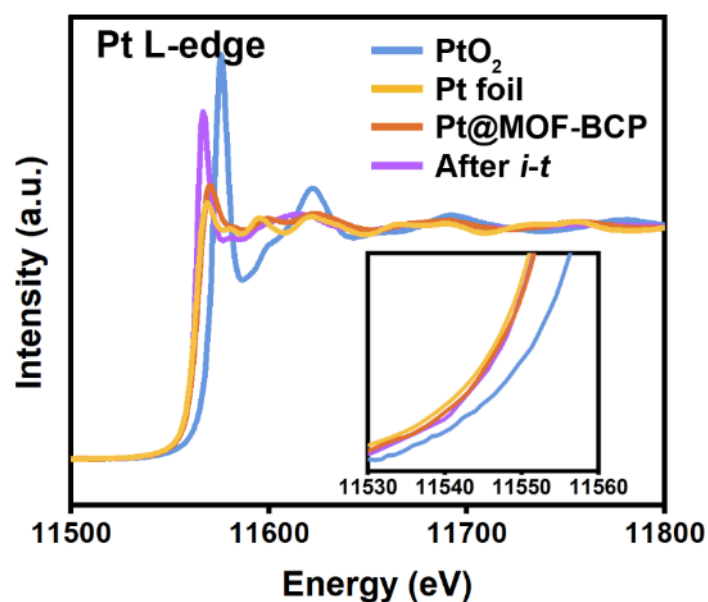


Figure S86. Normalized Pt L-edge XANES spectra of Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

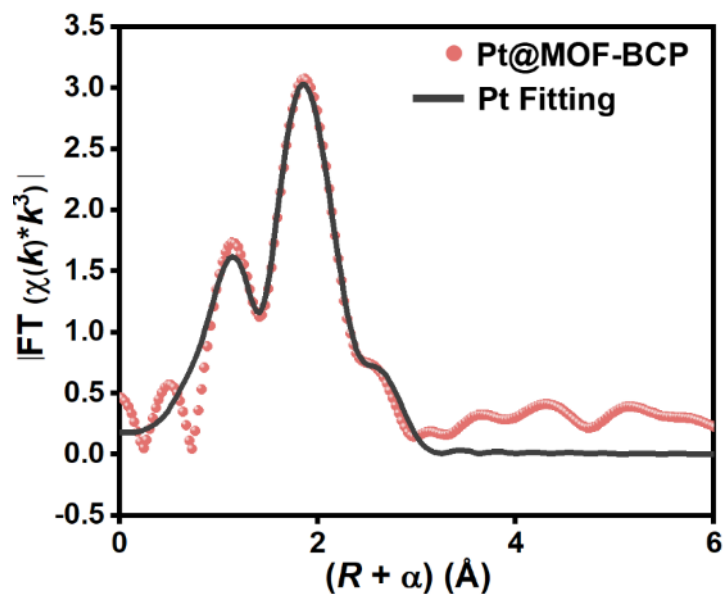


Figure S87. EXAFS fitting results of Pt for Pt@MOF-BCP after HER test in 0.5 M H₂SO₄.

Table S9. EXAFS fitting parameters at Pt L-edge for Pt@MOF-BCP sample after HER test in 0.5 M H₂SO₄. ($S_0^2 = 0.826$)

Sample	Scattering pair	R (Å)	* σ^2 (Å ²)	ΔE_0 (eV)	R factor
Pt@MOF-BCP	Pt-N	1.82 ± 0.09	0.018	-7.6	0.032
	Pt-Fe	2.50 ± 0.02	0.018		
	Pt-Pt	2.80 ± 0.15	0.016		

Data ranges: $3 \leq k \leq 12.2 \text{ Å}^{-1}$, $1 \leq R \leq 2.5 \text{ Å}$. The number of variable parameters is 7, out of a total of 8.5 independent points. R-factor for this fit is 3.2 %.

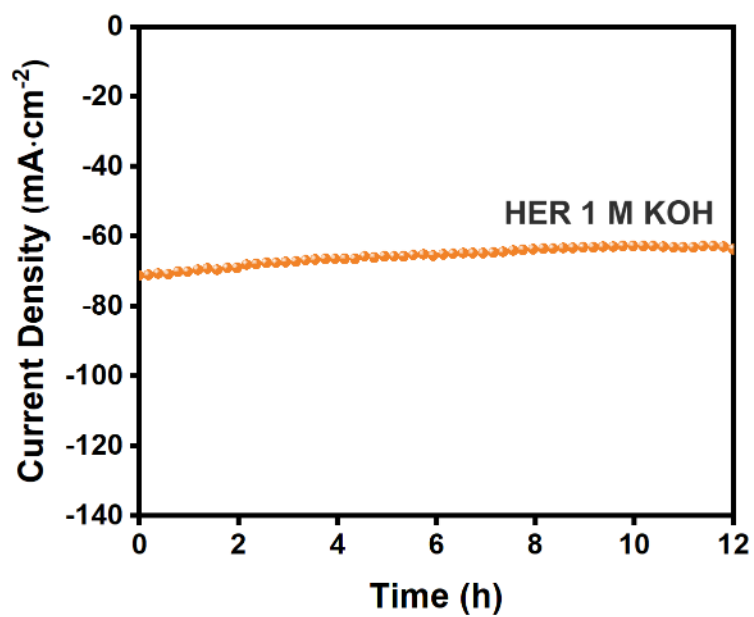


Figure S88. The *i-t* test for HER in 1 M KOH.

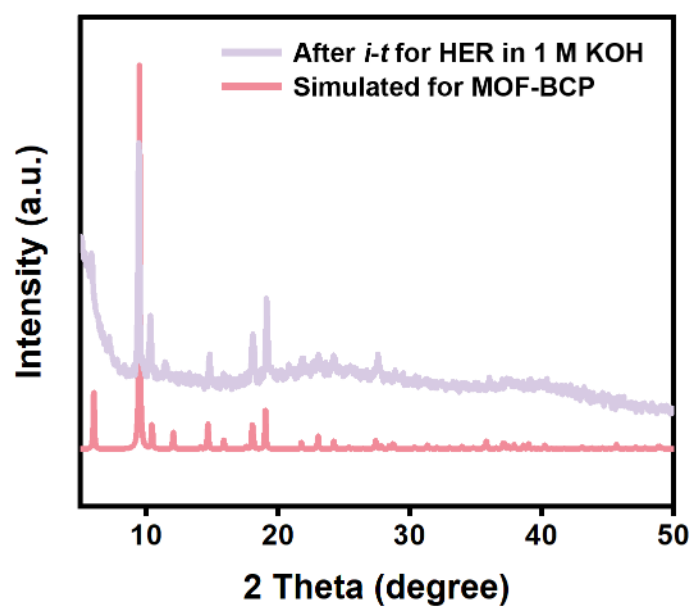


Figure S89. XRD pattern of Pt@MOF-BCP after HER test in 1 M KOH.

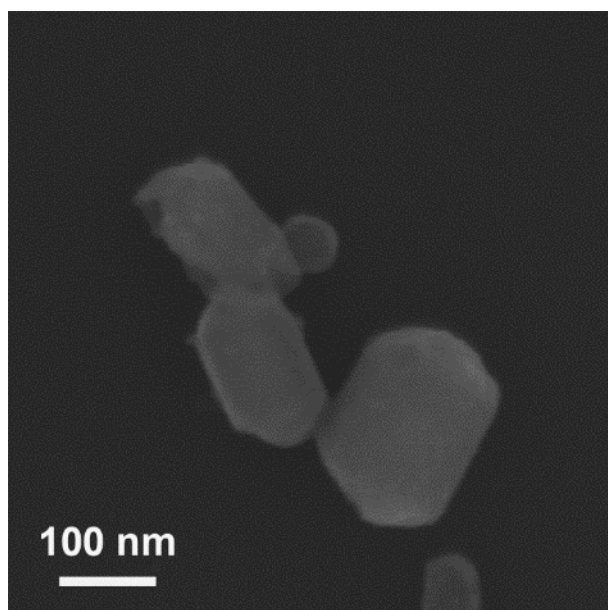


Figure S90. SEM image of Pt@MOF-BCP after HER in 1 M KOH.

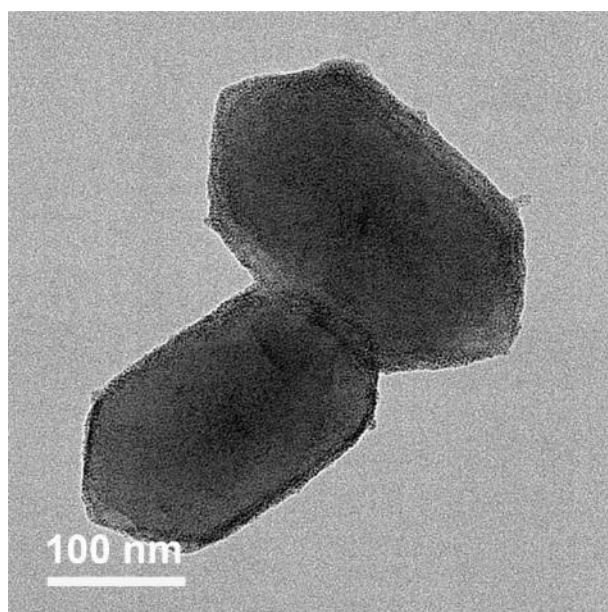


Figure S91. TEM image of Pt@MOF-BCP after HER in 1 M KOH.

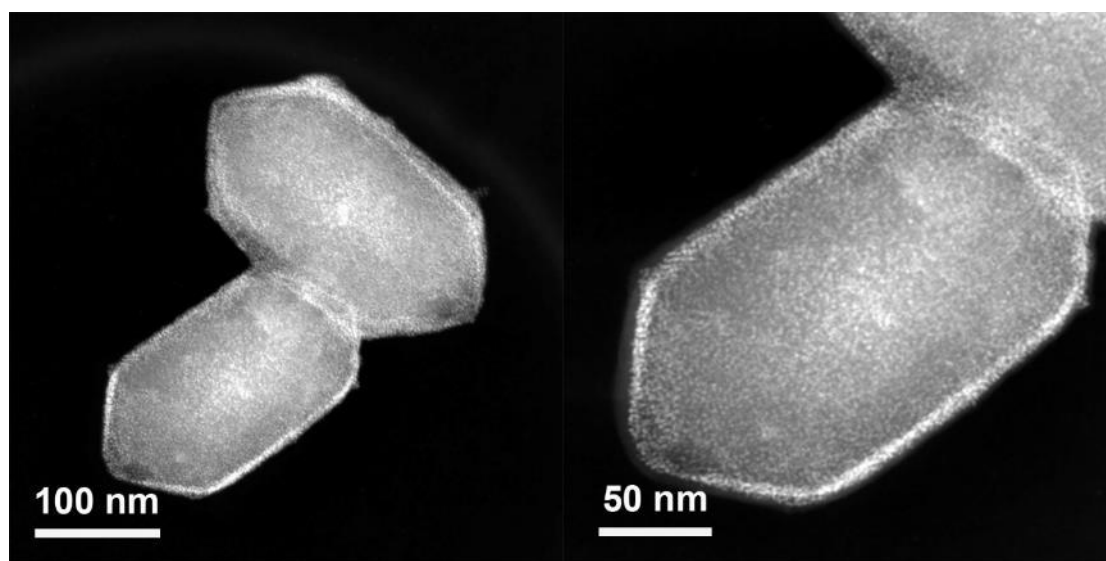


Figure S92. HAADF-STEM image of Pt@MOF-BCP after HER test in 1 M KOH.

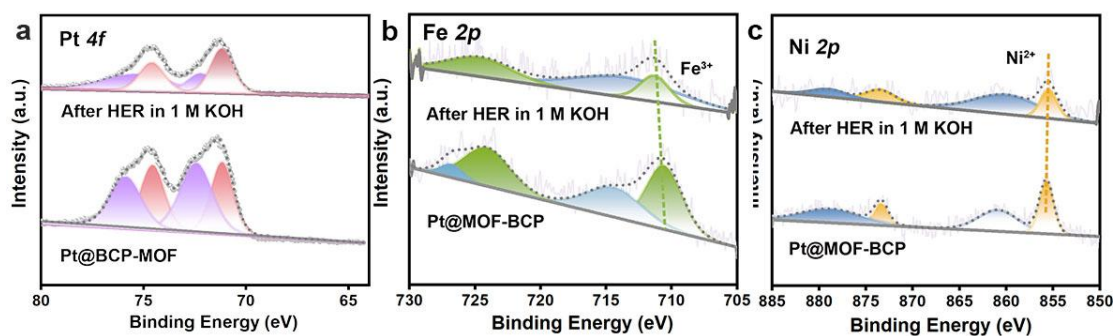


Figure S93. XPS spectra of (a) Pt 4f, (b) Fe 2p and (c) Ni 2p of Pt@MOF-BCP after HER test in 1 M KOH.

Note: Pt@MOF-BCP also exhibited an excellent stability after the HER durability test in 1 M KOH (Figure S88). The XRD, SEM, TEM and HAADF-STEM revealed that the original structure and morphology of Pt@MOF-BCP were well retained after HER. XPS analysis demonstrated that the valence state of Pt clusters remained largely unchanged, confirming the electronic structure stability of the catalyst under alkaline HER conditions (Figures S89-S93).

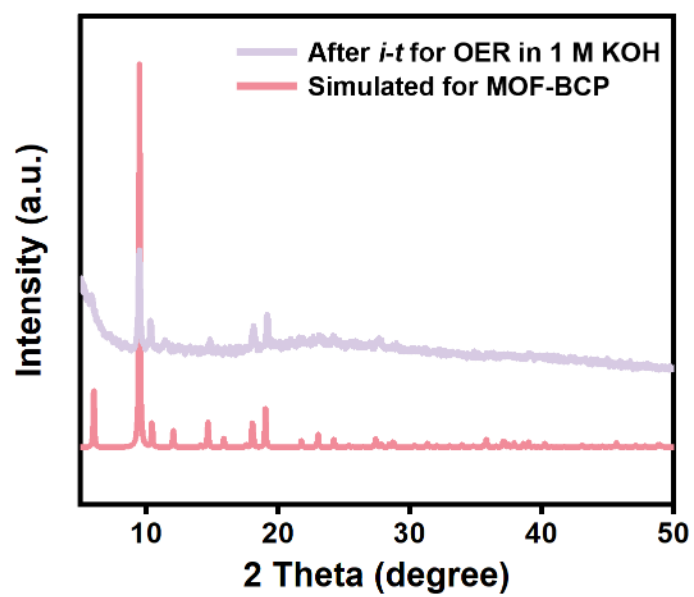


Figure S94. XRD pattern of Pt@MOF-BCP after OER test in 1 M KOH.

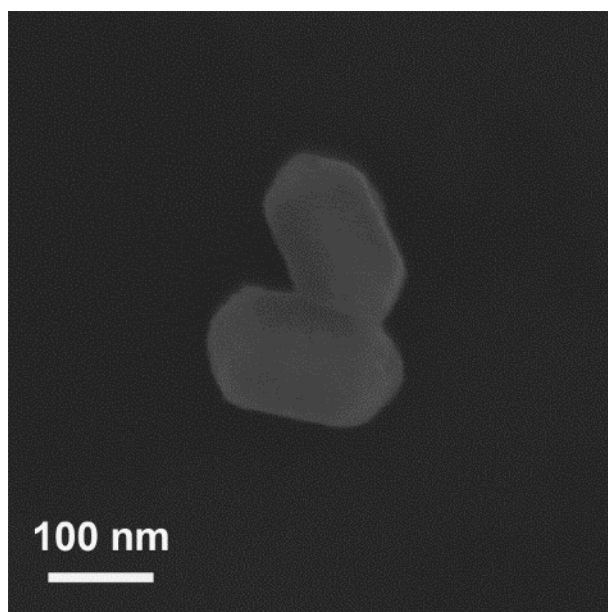


Figure S95. SEM image of Pt@MOF-BCP for OER in 1 M KOH.

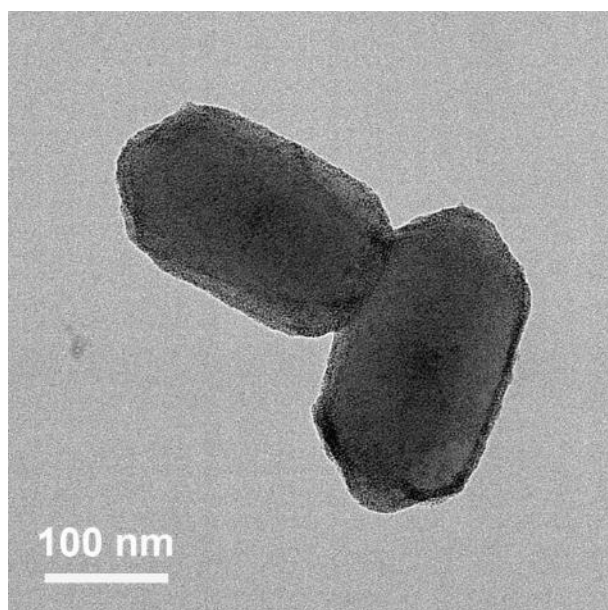


Figure S96. TEM image of Pt@MOF-BCP for OER in 1 M KOH.

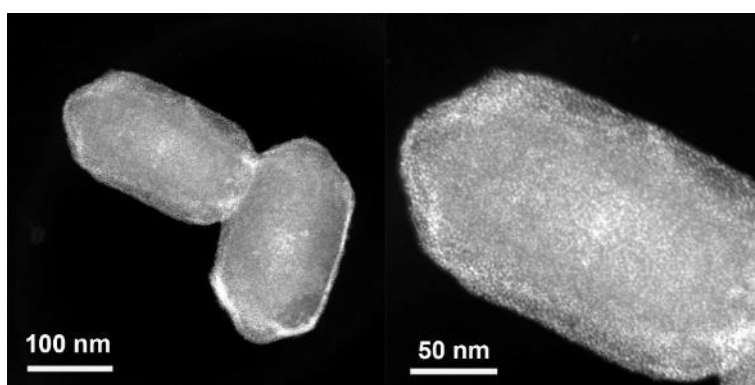


Figure S97. HAADF-STEM image of Pt@MOF-BCP for OER in 1 M KOH.

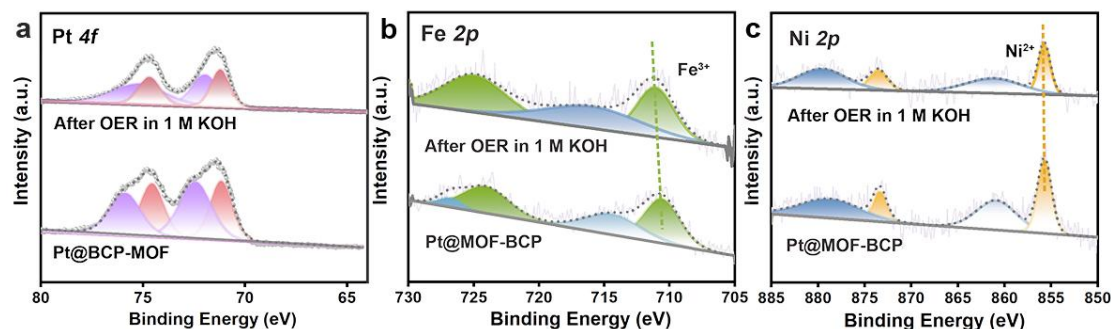


Figure S98. XPS spectra of (a) Pt 4*f*, (b) Fe 2*p* and (c) Ni 2*p* of Pt@MOF-BCP after OER test in 1 M KOH.

Note: XRD pattern, along with SEM, TEM and HAADF-STEM images of Pt@MOF-BCP, showed no evidence of Pt aggregation or MOF structural degradation after OER testing in 1.0 M KOH (Figures S94-S97). The Pt 4*f* XPS spectra exhibited only slight changes. In addition, the appearance of partially high-valence metal centers (Fe/Ni) in Pt@MOF-BCP after OER process is attributed to the inevitable oxidation process that occurs under alkaline OER conditions (Figure S98).

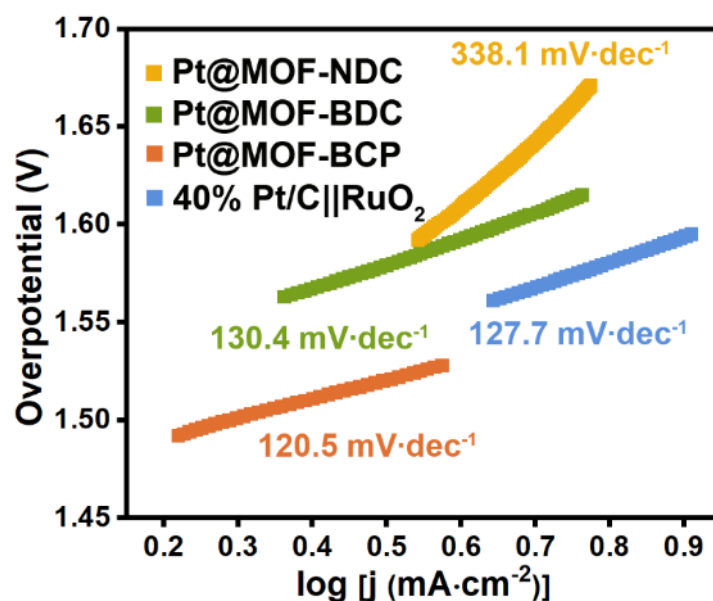


Figure S99. Tafel slopes of Pt@MOFs and 40% Pt/C||RuO₂ for overall water splitting in 1M KOH.

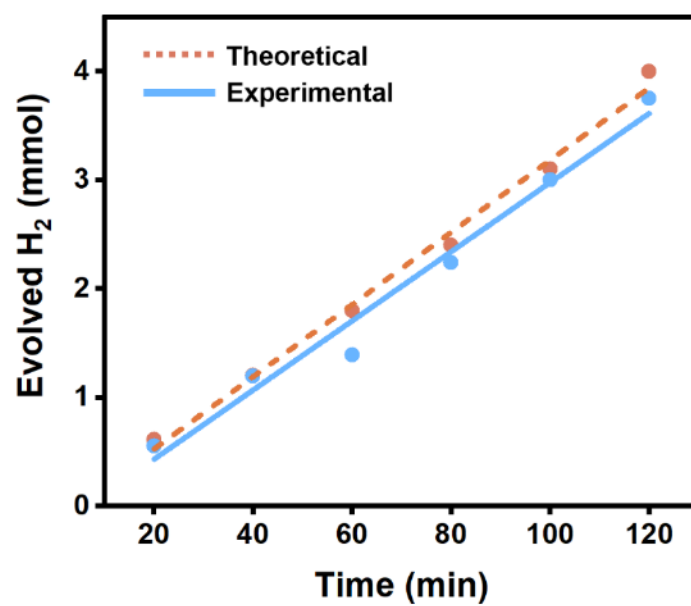


Figure S100. Theoretical and experimental amounts of H₂ evolved during the overall water splitting.

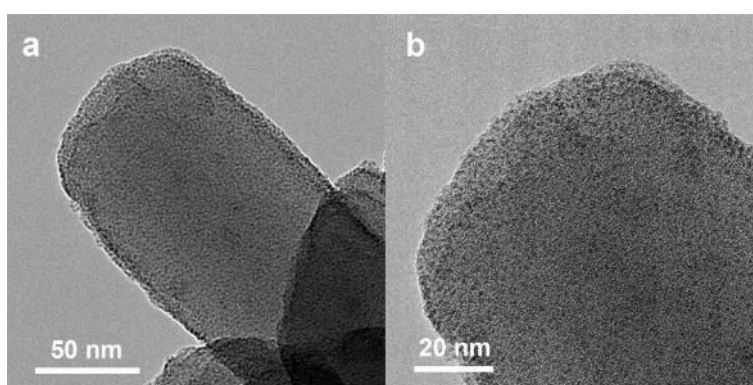


Figure S101. TEM image after *i-t* test of Pt@MOF-BCP for overall water splitting.

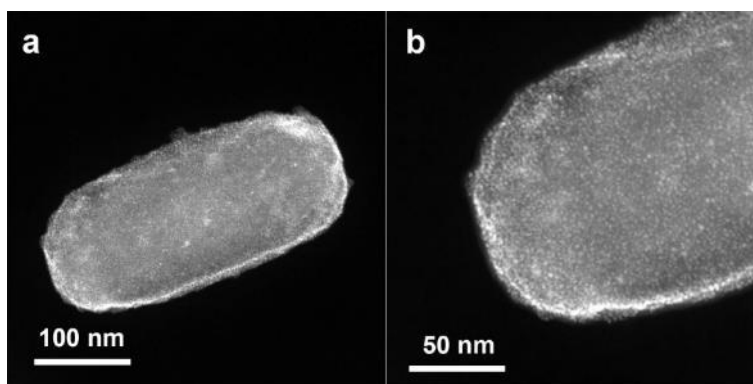


Figure S102. HAADF-STEM image after *i-t* test of Pt@MOF-BCP for overall water splitting.